

Turbulent natural convection in isothermally heated open-ended building shafts: Numerical analysis of ventilation performance

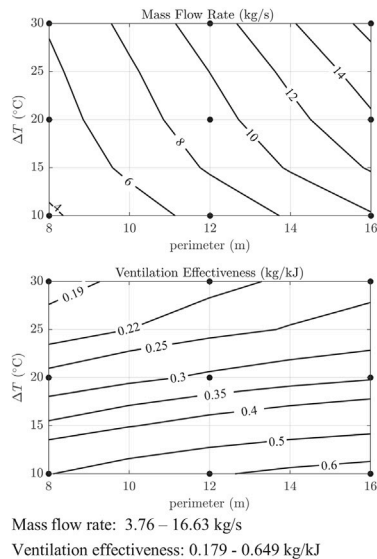
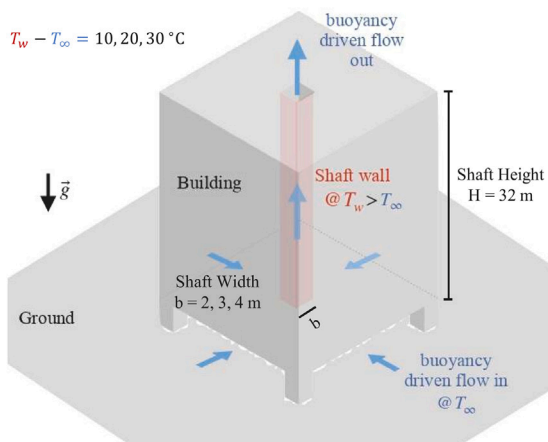
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GRAPHICAL ABSTRACT

We investigate buoyancy-driven ventilation in open-ended building shafts whose walls are isothermally heated, aiming to quantify removal of warm, polluted air from urban canyons.



ARTICLE INFO

Keywords:

Natural convection
Building shaft
Passive ventilation
Isothermal heating

ABSTRACT

We investigate buoyancy-driven ventilation in open-ended building shafts whose walls are isothermally heated, aiming to quantify passive removal of warm, polluted air from pedestrian-level urban canyons. Three-dimensional RANS CFD simulations (SST $k-\omega$) are carried out for a 36 m building containing a central shaft of height $H = 32$ m with square cross-sections $b = 2, 3$, and 4 m and wall-ambient temperature differences $\Delta T = 10, 20, 30^\circ\text{C}$, corresponding

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<https://doi.org/10.1016/j.csite.2025.107553>

Received 9 October 2025; Received in revised form 28 November 2025; Accepted 12 December 2025

Available online 18 December 2025

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Nusselt number
Mass flow rate
Ventilation effectiveness
Rayleigh number

to $Ra \approx 2.9 \times 10^{13} - 7.6 \times 10^{13}$ (turbulent regime). A sensitivity study comparing far-field boundaries modeled as stationary walls versus zero gauge pressure outlets shows that the resulting mass flow rates differ by less than 10 %. The flow remains developing along the shaft and exhibits inlet separation with a delayed reattachment in wider shafts; three zones are identified: recirculation, core-flow, and boundary-dominated. Predicted mass flow rates span 3.76 – 16.63 kg/s, with ventilation effectiveness (mass flow per unit heat input) 0.179–0.645 kg/kJ; increasing b improves effectiveness, whereas increasing ΔT enhances heat transfer faster than mass flow, reducing effectiveness. Partial validation against Churchill–Chu correlations for an isothermal vertical plate shows average Nusselt numbers within ≤ 15 % over most of the plate length. Two simplified models (a stack model with losses and a 1-D model) capture trends but underpredict 3-D effects tied to corner dynamics and entry recirculation; accuracy improves when the stack model uses a friction factor inferred from wall shear. From the CFD data, we propose building-scale correlations for the average Nusselt number, effective friction factor, and entry length as functions of Rayleigh number and aspect ratio b/H , to guide preliminary design of passively ventilated shafts in urban settings.

1. Introduction

The detrimental impacts of urbanization on human health and the environment have been the subject of numerous investigations. Degraded outdoor thermal conditions and increased levels of pollutant concentration have been associated with increased rates of mortality and morbidity, increased energy consumption in buildings, and decreased efficiency of power plants among others [1]. In 2018, the United Nations estimated that around 55% of the world population resided in urban areas, and this number is projected to increase to 68% by 2050 [2]. Consequently, it is essential to abate urban overheating and disperse pollutants at the pedestrian level. Introducing open vertical shafts in buildings helps increase wind flow in urban contexts by making the built environment more porous. This paper reports on a numerical study of natural convection in a building shaft. Such a configuration can be potentially employed to passively ventilate urban street canyons, leading to improved air quality and outdoor conditions.

Heat transfer by natural convection is encountered in a wide range of engineering applications such as cooling of electronic equipment [3], passive heating of buildings [4], and solar collectors [5] to mention a few. The different factors affecting internal natural convection heat transfer have been extensively studied with special attention given to natural convection in enclosures, where the flow is commonly established via a temperature difference between the inflow and the enclosure walls. These include incoming flow with ambient conditions with the enclosure walls held at a higher temperature (realized as an isothermal condition [6] or a heat flow condition [7]), or preheating the flow prior to entry into the enclosure [8,9]. Impact of the enclosure geometry and fluid type were reviewed by Rahimi et al. in [10]. Adding an obstacle to the flow in the enclosure was reviewed by Pandey et al. [11]. A review of the various numerical methods (finite volume, finite difference, and Lattice Boltzmann), and experimental methods used to study internal flow fields established by natural convection can be found in Miroshnichenko and Sheremet [12].

Table 1 lists numerical and experimental studies, in two- and three-dimensional domains, of laminar and turbulent buoyancy-driven flows in vertical open-ended channels.

When the aspect ratio of the channel cross-section is sufficiently large and the inflow and boundary conditions permit, the flow can be modeled as two-dimensional, which allows for predicting of the flow field using cost-effective numerical solution approaches. In this context, two-dimensional laminar buoyancy driven flows were numerically studied in [6,7,13–15] using finite difference [13–15] and finite volume [6,7] methods. These studies unraveled key features of these flows. Bodoia and Osterle [13] studied the buoyancy-driven flow induced between symmetrically heated parallel plates and concluded that fully developed flow conditions can practically be attained for highly viscous flows. Naylor et al. [14] applied a Jeffery–Hamel based inlet boundary condition for a more realistic representation of the conditions at the inlet. The results revealed flow separation at the entrance of the channel when Grashof number (Gr) is sufficiently large. Laminar natural convection flows in symmetrically heated vertical channels with curved surfaces were numerically investigated for isothermal [6] and isoflux conditions [7]. For the isothermal case, the results showed that the Nusselt Number (Nu) profile decreases with height, which was attributed to the development of the thermal boundary layer that acts as a thermal resistance to reduce heat transfer from the wall to the ambient. Moreover, clustering of isotherms has been observed at high values of Rayleigh number (Ra), which is associated with the dominance of the convection effects over the conduction effects. Boudebos et al. [15] explored the impacts of the geometric configuration and Ra on the characteristics of free convection by symmetrically heating a hyperbolic duct. The results showed that, in laminar flows, as Ra increases, the hydrodynamics governing the flow become less influenced by the geometric configuration.

Turbulent natural convection flows in two-dimensional domains have been numerically studied for symmetrically [16,17] and asymmetrically [18,19] heated vertical channels of constant rectangular cross-section. For the symmetrically heated case, Badr et al. [16] investigated the impacts of the modified Rayleigh number (Ra_m) and the aspect ratio of the channel on the flow, subject to isothermal and iso-flux conditions. The authors also presented Nusselt number correlations in terms of Ra_m and the aspect ratio of the channel. In their numerical analysis of turbulent free convection flows in a solar chimney, Zamora & Kaiser [17] employed the $k - \omega$ turbulence closure model. The optimum values of the wall-to-wall spacing that maximizes the mass flow rate were found to be higher than those that maximize Nu . In their study of the flow induced by asymmetric heating, Fedorov and Viskanta [19] reported that the $k - \epsilon$ turbulence model coupled with wall-damping functions can predict the thermal behavior of a free-convection

Table 1
Studies on natural convection in open-ended vertical channels.

		Laminar	Turbulent
2D	Numerical	[6,7,13–15]	[16–19]
3D	Numerical	[20]	[21]
	Experimental	[20]	[21–23]

flow generated between a heated plate and an adiabatic plate in an infinite volume of fluid. However, accurate prediction required tuning of the turbulence intensity of the flow at the inlet, underscoring the need for further experimental work to validate the height at which the transition from a laminar to a turbulent regime occurs.

Even though natural convection heat transfer in two-dimensional domains has been thoroughly studied, studies of buoyancy-driven air flows in three-dimensional vertical channels are scarce. Free convection flows in vertical channels, consisting of two parallel vertical plates, have been experimentally (and numerically with 2D models) investigated for laminar [20] and turbulent [21–23] flow regimes. Straatman et al. [20] conducted a numerical and experimental study to quantify the enhancement in the overall heat transfer as a result of adding adiabatic extensions to the outlet of an isothermally symmetrically heated vertical channel. The authors presented a correlation relating the Nu number to Ra in addition to the dimensions of the channel and the extension. Miyamoto et al. [22] experimentally investigated the thermal and hydrodynamic characteristics of a buoyancy-driven turbulent flow in a vertical channel, where one wall is insulated and the other is maintained at a constant heat flux condition. Habib et al. [23] presented experimental measurements of velocity profiles at three horizontal cross-sections between symmetrically heated and anti-symmetrically heated vertical parallel plates. For the symmetrically heated case, the flow did not reach full development, and the maximum mean velocity, which was attained near the channel walls, increased at higher elevations. Roelveld et al. [21] experimentally and numerically analyzed the turbulent buoyant flow characteristics in an asymmetrically heated vertical channel. Several cases were considered where the temperature difference between the hot wall and the ambient is not necessarily equal to the temperature difference between the ambient and the cold wall. Results revealed the presence of flow instabilities at relatively low modified Rayleigh numbers (Ra_m) for the case of opposing buoyancy forces.

In three-dimensional vertical ducts, studies of buoyancy-driven air flow are largely limited to laminar flows. Ramakrishna et al. [24] performed a computational analysis using the finite difference scheme to analyze the characteristics of the buoyancy-driven laminar flow induced in a symmetrically heated three-dimensional duct. Moutsoglou & Park [25] conducted numerical studies to compare heat transfer rates and flow rates between a three-dimensional duct and a two-dimensional vertical channel. For both geometries, a free-convection laminar flow is established by uniformly heating one wall and insulating the others. The results revealed that the two-dimensional model neglects the restraining effects of the lateral edges, and consequently overestimates Nu and velocities when compared to the three-dimensional model. Moreover, the authors reported that for sufficiently large spacing between the lateral edges, the three-dimensional effects can be neglected.

To the authors knowledge, there are no studies that report on three-dimensional buoyancy-driven turbulent flows in vertical ducts at the building or street canyon scale. Such studies are necessary to assess methods of ventilating street canyons to mitigate the heat island and low air quality in urban environments. This study employs a computational approach to (1) explore the characteristics of turbulent three-dimensional natural convection air flow in an isothermally heated open-ended building shaft, (2) investigate the impact of the shaft width and the temperature difference between the walls and the ambient (ΔT) on the mass flow rate of air and the ventilation capacity of the shaft, and (3) propose a building-scale correlation for the average Nusselt number, $\widetilde{Nu}$, friction factor $\tilde{f}$, and reattachment length ℓ_1 , as a function of Rayleigh number (Ra) and the shaft aspect ratio. These correlations provide a useful basis for the preliminary design of buoyancy-driven ventilation shafts in similar applications.

The remainder of this article is organized as follows. Section 2 begins with a brief description of the building geometry, followed by a presentation of the equations governing the conservation of mass, momentum, energy, and turbulence closure quantities, expressed in dimensionless form for buoyancy-driven turbulent flows. The numerical setup and computational grid are then presented, along with a partial validation of the results. Section 2 is concluded with presenting simplified models for a buoyancy-driven flow in a vertical shaft. In Section 3, the dependence of the mass flow rate and ventilation effectiveness on the operating conditions is discussed, followed by a comparison between the results of the numerical simulations and the simplified models. The three-dimensional characteristics of the flow are then explored by inspecting the streamline at the entry, axial and cross sectional velocity profiles, and isotherm cross sectional contours. Finally, correlations for the $\widetilde{Nu}$ and $\tilde{f}$ in terms of (Ra) and the aspect ratio are presented.

2. Method

2.1. Problem description - Geometry

In this study, the buoyancy-driven flow up the vertical shaft is established by keeping the shaft walls at a temperature higher than that of the surrounding. This difference in temperature creates a heat flux that heats air in the vicinity of the walls. As a result, the lower air density gives rise to the buoyancy force driving the flow. As shown in Fig. 1, the outdoor air surrounding the building is naturally convected up in the vertical shaft. The net effect is passively ventilating the area around the building, thus improving

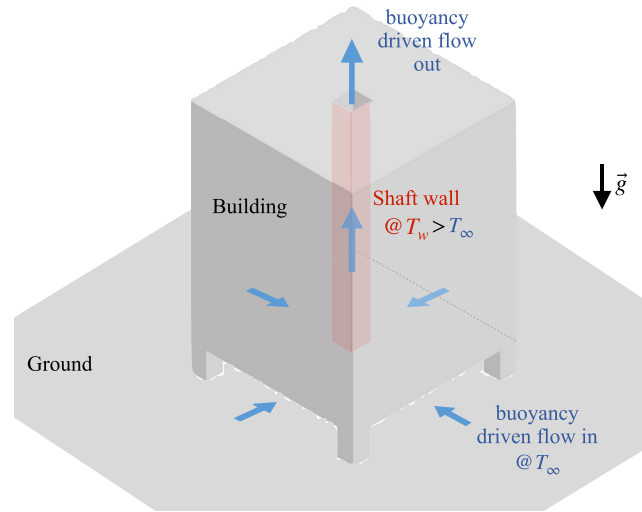


Fig. 1. Ambient air, at T_∞ is driven into the vertical by buoyancy arising from the higher shaft wall temperature, T_w .

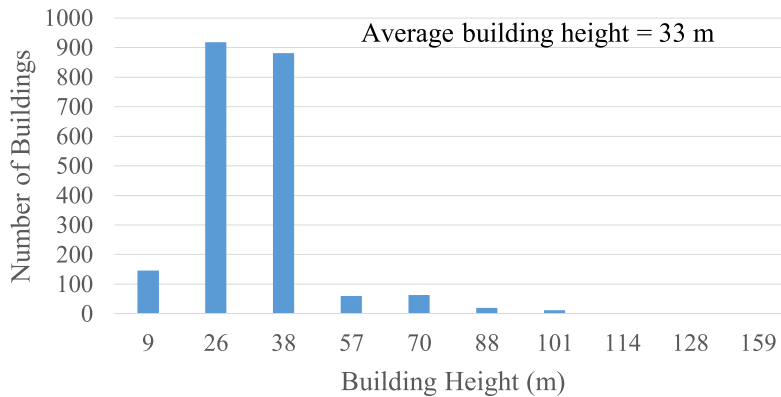


Fig. 2. Buildings heights statistics in Beirut city for the period 2000–2024.

the outdoor comfort conditions. The building height is selected to be 36 m, which is in line with recent trends in the construction sector in Beirut. Based on building statistics collected by the Beirut Urban Lab, Fig. 2 shows a histogram of the buildings constructed since 2000, divided into ten bins according to their height. We observe in this figure that the average height is around 33 m. The footprint of the building is chosen to be square with an area of 625 m², which is typical for four two- or three-bedroom apartments. Noting that while the size and shape of the building footprint does not considerably affect the ventilation capacity at the building scale, the building footprint and spacing between buildings are key parameters when studying pollutant dispersion at a larger urban scale involving multiple buildings or blocks.

Fig. 3 shows a schematic of a building of height 36 m whose footprint is a square of side 25 m. The ground floor, with a height of 4 m, is open to the surrounding environment, and an open-ended square shaft of height (H) 32 m runs through the center of the building. The following widths of the shaft cross section are considered: 2 m, 3 m, and 4 m. The computational domain, shown in Fig. 3, is created by subtracting the volume of the building from the volume of the fluid and includes both the air surrounding the building and the air inside the shaft. Due to symmetry, the computational domain consists of one-fourth of the fluid domain. The side boundaries of the fluid domain are taken 150 m away from the center of the shaft, while the upper boundary is set 284 m away from the roof of the building.

The buoyant flow established for the studied ranges of shaft geometry and wall temperature is turbulent. This is consistent with the corresponding range of Rayleigh number values that characterize the numerically predicted flow, as presented in Section 2.2. The conservation equations for mass, momentum, and energy governing turbulent, thermally-driven buoyant flows are expressed in Eqs. (1)–(3), respectively. For the turbulence closure problem, we employ the SST $k - \omega$ turbulence model, which has been used by some studies to model turbulent buoyancy-driven flows [26,27]. The transport equations of the turbulent kinetic energy (k) and specific turbulent dissipation rate (ω) are, respectively, expressed in Eqs. (4) and (5).

$$\nabla \cdot (\rho \vec{u}) = 0 \quad (1)$$

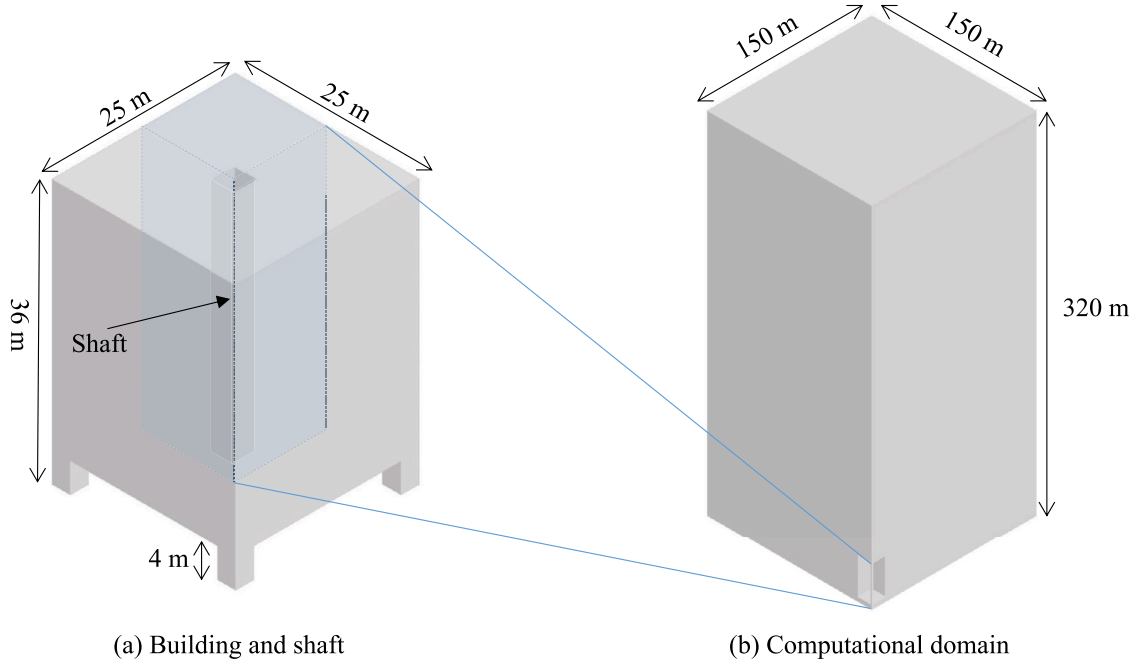


Fig. 3. Geometry and computational domain.

Table 2

Dimensionless numbers, $\Delta T = T_w - T_\infty$, $L = H$.

$\nabla^* = L\nabla$	$\rho^* = \frac{\rho}{\rho_\infty}$	$\vec{u}^* = \frac{\vec{u}}{\sqrt{gL\beta\Delta T}}$	$p^* = \frac{p_{eff}}{\rho_\infty RT_\infty}$
$\hat{z} = -\frac{\vec{z}}{g}$	$\mu^* = \frac{\mu}{\mu_\infty}$	$\mu_t^* = \frac{\mu_t}{\mu_\infty}$	$c_p^* = \frac{c_p}{c_{p,\infty}}$
$T^* = \frac{T - T_\infty}{\Delta T}$	$\lambda^* = \frac{\lambda}{\lambda_\infty}$	$\lambda_t^* = \frac{\lambda_t}{\lambda_\infty}$	$\Phi^* = \frac{\Phi L}{g}$
$k^* = \frac{k}{gL\beta\Delta T}$	$\Gamma_k^* = \frac{\Gamma_k}{\mu_\infty g}$	$S_k^* = \frac{S_k L}{\mu_\infty g}$	$G_k^* = \frac{G_k L}{\mu_\infty g\beta\Delta T}$
$Y_k^* = \frac{Y_k L}{\rho_\infty (gL\beta\Delta T)^{\frac{1}{2}}}$	$G_b^* = \frac{G_b L Pr_t}{\mu_\infty g}$	$w^* = \frac{w L}{\sqrt{gL\beta\Delta T}}$	$\Gamma_\omega^* = \frac{\Gamma_\omega}{\mu_\infty}$
$S_\omega^* = \frac{S_\omega L}{\rho_\infty g}$	$G_\omega^* = \frac{G_\omega L}{\rho_\infty g\beta\Delta T}$	$Y_\omega^* = \frac{Y_\omega L}{\rho_\infty g\beta\Delta T}$	$G_{\omega b}^* = \frac{G_{\omega b} L \sqrt{Gr Pr_t}}{\rho_\infty g}$
$D_\omega^* = \frac{D_\omega L}{\rho_\infty g\beta\Delta T}$	$Gr = \frac{\rho^3 g \beta L^3 \Delta T}{\mu_\infty^2}$	$Pr = \frac{\mu c_p}{\lambda}$	$Ra = Gr Pr$
$Nu = \frac{hL}{\lambda}$	$\hat{z} = \frac{z}{L}$		

$$\nabla \cdot (\rho \vec{u} \vec{u}) = -\nabla (p_{eff} - p_\infty) + (\rho - \rho_\infty) \vec{g} + \nabla \cdot ((\mu_{eff}) \nabla \vec{u}) + \nabla \cdot (\mu_t (\nabla \vec{u})^T) \quad (2)$$

$$\nabla \cdot (\rho c_p T \vec{u}) = \nabla \cdot (\lambda_{eff} \nabla T) + \mu \Phi \quad (3)$$

$$\nabla \cdot (\rho \vec{u} k) = \nabla \cdot (\Gamma_k \nabla k) + G_k - Y_k + S_k + G_b \quad (4)$$

$$\nabla \cdot (\rho \vec{u} \omega) = \nabla \cdot (\Gamma_\omega \nabla \omega) + G_\omega - Y_\omega + D_\omega + S_\omega + G_{b\omega} \quad (5)$$

where $p_{eff} = p + \frac{2}{3} \rho k$, $\mu_{eff} = \mu + \mu_t$, $\lambda_{eff} = \lambda + \lambda_t$, g (m/s²) is the gravitational acceleration, Φ (1/s²) is the viscous dissipation term, ρ (kg/m³) is the density, μ (kg/ms) is the dynamic viscosity, c_p (kJ/kg K) and λ (W/mK) are the constant pressure specific heat and thermal conductivity of air, respectively. The coefficients of the turbulence model employed [28–31] are presented in Appendix A. Section 2.2 presents a discussion on the flow regime and a justification for using the SST $k - \omega$ turbulence model.

Using the dimensionless variables outlined in Table 2, the non-dimensional form of Eqs. (1)–(5) is presented in Eqs. (6)–(10). Since the flow is a developing flow, the shaft height is selected as the characteristic length, providing a consistent reference scale for the analysis. Since the flow in this study is driven by buoyancy, the characteristic velocity is derived as a function of the shaft height and the buoyancy force, which depends on gravitational acceleration and the temperature gradient between the wall surfaces and the ambient environment.

$$\nabla^* \cdot (\rho^* \vec{u}^*) = 0 \quad (6)$$

Table 3
Values of Rayleigh for the cases considered.

H (m)	Ra		
	$\Delta T = 10$ °C	20 °C	30 °C
32	2.9×10^{13}	5.5×10^{13}	7.6×10^{13}

$$\nabla^* \cdot (\rho^* \vec{u}^* \vec{u}^*) = -\frac{RT_\infty}{gL\beta\Delta T} \nabla^* \cdot (p^*) + \frac{1}{\beta\Delta T} (1 - \rho^*) \hat{z} + \frac{1}{\sqrt{Gr}} \nabla^* \cdot (\mu_{eff}^* \nabla^* \vec{u}^*) + \frac{1}{\sqrt{Gr}} \nabla^* \cdot (\mu_t^* (\nabla^* \vec{u}^*)^T) \quad (7)$$

$$\nabla^* \cdot (\rho^* T^* \vec{u}^*) + \frac{T_\infty}{T_w - T_\infty} \nabla^* \cdot (\rho^* \vec{u}^*) = \frac{1}{Pr\sqrt{Gr}} \nabla^* \cdot ((1 + \lambda_t^*) \nabla^* T^*) + \frac{gL}{c_p \sqrt{Gr} (T_w - T_\infty)} \Phi^* \quad (8)$$

$$\nabla^* \cdot (\rho^* \vec{u}^* k^*) = \frac{1}{\sqrt{Gr}} \nabla^* \cdot (\Gamma_k^* \nabla^* k^*) + \frac{1}{\sqrt{Gr}} G_k^* - Y_k^* + \frac{1}{\beta\Delta T \sqrt{Gr}} S_k^* + \frac{1}{\beta\Delta T Pr_t \sqrt{Gr}} G_b^* \quad (9)$$

$$\nabla^* \cdot (\rho^* \vec{u}^* \omega^*) = \frac{1}{\sqrt{Gr}} \nabla^* \cdot (\Gamma_\omega^* \nabla^* \omega^*) + G_\omega^* - Y_\omega^* + D_\omega^* + \frac{1}{\beta\Delta T} S_\omega^* + \frac{1}{\beta\Delta T \sqrt{Gr} Pr_t} G_{\omega b}^* \quad (10)$$

where β (1/K) is the thermal expansion coefficient, T_w (K) is the wall temperature, T_∞ (K) is the ambient temperature, and L (m) is the characteristic length.

2.2. Numerical setup and simulation

The numerical solutions are generated using ANSYS Fluent [32]. Air is modeled as an ideal gas of constant specific heat, dynamic viscosity, and thermal conductivity. The ideal gas model is used since it provides a more accurate representation of density variations than the Boussinesq approximation, which is a linearization of the density–temperature relation and is valid only for small temperature differences satisfying $\beta(T - T_\infty) \ll 1$ [32]. To confirm that compressibility effects are negligible, an additional simulation was conducted for one case, where we modeled air as an incompressible ideal gas. The predicted mass flow rate and wall heat transfer remained unchanged, indicating weak compressibility. Additionally, all simulations were run using a pressure-based solver, which is suitable for incompressible and moderately compressible flows. The coupled algorithm is selected to resolve the pressure–velocity coupling [33–41] and the body force weighted average scheme is adopted for pressure interpolation. Moreover, the 2nd order upwind scheme [42] is used for the discretization of the convective fluxes in the momentum, energy, and turbulence transport equations. Finally, spatial gradients are calculated via the least square cell based method.

The importance of buoyancy-driven forces relative to opposing viscous and thermal dissipative effects is quantified by the Rayleigh number (Ra), given in Table 2. According to Table 3, Ra is larger than 10^9 , which is the critical Ra for the buoyant flow over a vertical plate, suggesting that the flow is turbulent [43]. This is confirmed by the fact that the value of Reynolds number based on the numerically predicted velocity of the buoyancy-driven flow exceeds 100,000 for all the cases considered.

Numerical simulation of buoyancy-driven turbulent flows is challenging because of the strong coupling between thermal and momentum fields, leading to two key difficulties: (i) selecting an appropriate turbulence model and (ii) specifying the boundary conditions. A widely used approach in turbulence modeling involves Reynolds averaging of the Navier–Stokes (RANS) equations, where Reynolds stresses and turbulent heat fluxes appear as unknowns requiring closure. Commercial computational fluid dynamics (CFD) tools typically relate these terms to mean flow properties through turbulent viscosity (μ_t) and thermal conductivity (λ_t). This reduces the problem to computing μ_t by solving transport equations for the turbulent kinetic energy k and either its dissipation rate ε or specific dissipation rate ω . Since these two-equation models embed empirical parameters, their generalization to all flow conditions, especially buoyancy-driven flows, is limited. Hanjalić [44] extensively discussed the challenges faced by two-equation turbulence models in accurately predicting natural convection flows and proposed an algebraic flux model that solves an additional transport equation for temperature variance. However, improvements in handling buoyancy have been incorporated into some two-equation models including the SST $k - \omega$ model, which yielded reliable predictions for this class of problems.

To assess the performance of a turbulence model in natural convection flows, it is common practice to test it against canonical benchmark problems, such as a differentially heated cavity and a Rayleigh–Bénard Convection (RBC), for which experimental or Direct Numerical Simulation data exist. For the differentially heated cavity, Wu and Lei [27] reported that the SST $k - \omega$ model exhibited a superior overall performance compared with other two-equation turbulence models. With respect to RBC, although the flow has been extensively investigated experimentally and numerically, studies that compare the performance of turbulence models for this flow in three-dimensional transient simulations remain scarce. While performing a new RBC simulation is beyond the scope of this study, we plan, in a future study, to thoroughly investigate the sensitivity of this problem to the turbulence model using the findings of Xu et al. [45] as a reference case. Accordingly, the SST $k - \omega$ model is used to simulate turbulence effects in the present study.

The SST $k - \omega$ turbulence model incorporates a blending function in the turbulence equations allowing it to solve the standard $k - \omega$ equations near the walls for an accurate boundary layer resolution and the standard $k - \varepsilon$ equations far from the walls for their insensitivity to free-stream values. Hence, we expect that the SST $k - \omega$ model overcomes the limitations encountered when the standard $k - \varepsilon$ or standard $k - \omega$ models are used separately. Moreover, the model constrains the turbulent viscosity to satisfy Bradshaw's principle, which linearly relates the principal shear stress to the turbulent kinetic energy in the boundary

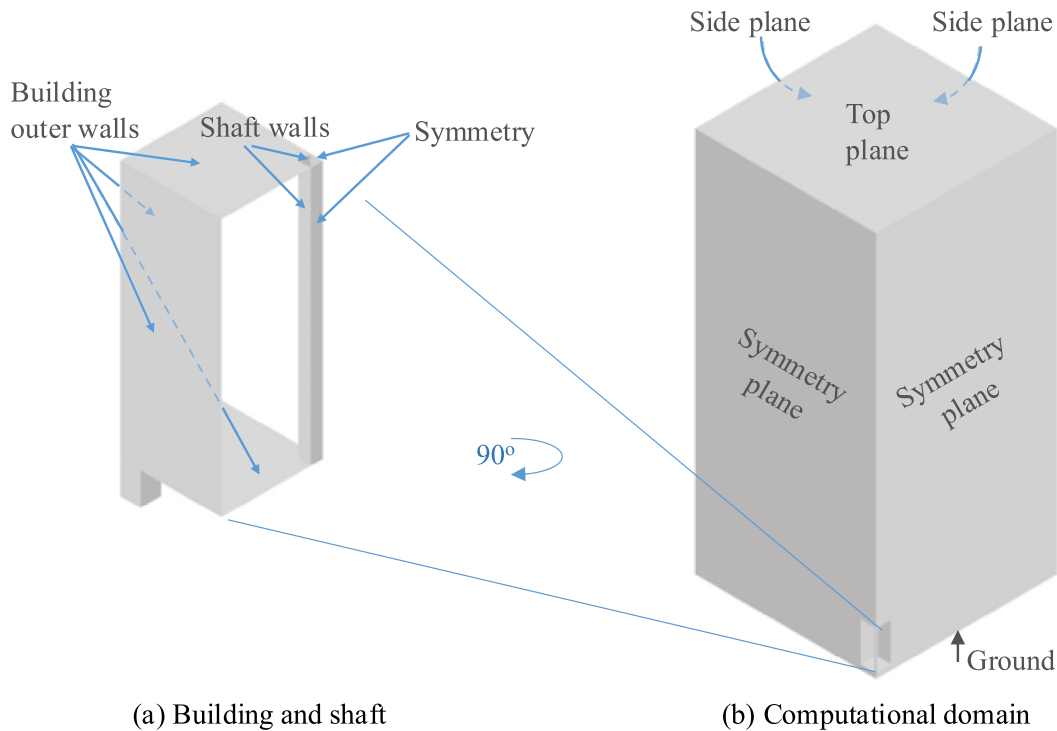


Fig. 4. Boundary conditions.

layer. The model assumes turbulent flow throughout, which is consistent with the observation that in a typical urban setting, the buoyancy-driven flow entering the building shaft is expected to be fully turbulent, owing to the upstream interactions with the built environment and vehicles. Implementing the SST $k - \omega$ model requires grid refinement near the walls, where significant gradients in physical properties occur. Details about the grid and its refinement near the walls are provided in Section 2.3 of this study.

In natural convection flows, the velocity at the inlet is not known beforehand, which poses challenges in assigning realistic boundary conditions. Some numerical studies extended the fluid domain below the entrance and assumed Jeffery–Hamel flow at the inlet and a pressure outlet condition at the outlet [6,7,14]. However, a Jeffery–Hamel solution describes a steady, two-dimensional, laminar radial flow between non-parallel walls [46]. As such, it is unsuitable for the present study, which involves turbulent, three-dimensional flow with a ground boundary beneath the shaft. Other studies extended the domain at the inlet, outlet, or both ends of the channel, applying pressure boundary conditions at the extended ends [47,48]. For such pressure boundary conditions, it is well documented that when the ideal gas law is used, the solution can be highly sensitive to the specified operating density [32]. To minimize artificial boundary influence and the sensitivity of the numerical model to the operating conditions, this study places the boundaries of the fluid domain sufficiently far from the shaft. These far-field planes — including the side boundaries, the top surface, and the ground (as shown in Fig. 4b) — are modeled as stationary walls maintained at a uniform temperature equal to that of the ambient, i.e. to $T_\infty = 25^\circ\text{C}$. The boundaries of the computational domain are depicted in Fig. 4 and summarized in Table 4.

Due to symmetry, only one-fourth of the fluid domain is modeled and symmetry boundary conditions are assigned to the surfaces that belong to the slicing planes as depicted in Fig. 4b. The outer walls of the building are presented in Fig. 4a and are assigned adiabatic boundary conditions. For each case of temperature difference considered, an isothermal condition is enforced on the shaft walls shown in Fig. 4a. The values of the shaft wall temperature considered in this study are 35°C , 45°C , and 55°C , corresponding to $\Delta T = 10^\circ\text{C}$, 20°C and 30°C .

Although the far-field boundaries were placed sufficiently far from the shaft, modeling them as stationary walls can still introduce artificial confinement effects. To examine the influence of boundary selection, we study the flow with the side and top boundaries defined as pressure outlets, assigning the gauge pressure (p_g) to zero and the backflow temperature (T_{br}) to the ambient value. We refer to the configuration using wall-type far-field boundaries as Scenario A, and the configuration using the zero gauge pressure-outlet boundaries as Scenario B.

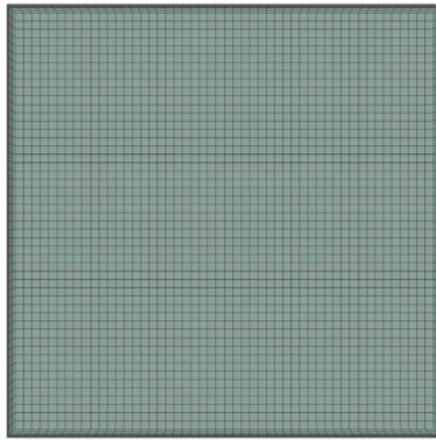
2.3. Meshing the computational domain

The computational domain is a hybrid grid comprising hexahedral and tetrahedral elements using the Mesher tool in ANSYS [49]. The critical regions in the domain are the shaft and the entrance region, which are discretized using an orthogonal hexahedral grid.

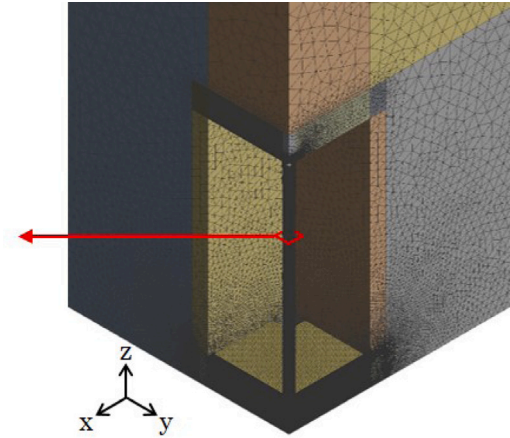
Table 4

Thermal and momentum boundary conditions. Planes of symmetry are shown in Fig. 4.

Surface	Momentum boundary conditions	Thermal boundary conditions
Top plane	Scenario A: $\vec{u} = \vec{0}$ (m/s)	Scenario A: $T = 25$ °C
Side plane	Scenario B: Pressure Outlet, $p_g = 0$ (kPa)	Scenario B: $T_{bf} = 25$ °C
Ground	$\vec{u} = \vec{0}$ (m/s)	$T = 25$ °C
Outer walls	$\vec{u} = \vec{0}$ (m/s)	$\dot{q} = 0$ (W/m ²)
Shaft walls	$\vec{u} = \vec{0}$ (m/s)	Case 1: $T_w = 35$ °C Case 2: $T_w = 45$ °C Case 3: $T_w = 55$ °C



(a) Grid in the shaft



(b) Grid in the computational domain

Fig. 5. The hybrid grid used to mesh the shaft and the building surroundings.**Table 5**

Mesh parameters and quality: b is the size of the shaft cross section, t_1 is the first inflation layer thickness, Δx is the grid size in the shaft cross section outside the inflation layer, N_E is the number of elements, ζ is the aspect ratio, and y^+_{max} is the maximum value of y^+ at the shaft wall.

b (m)	t_1 (mm)	Δx (mm)	N_E	ζ_{max}	y^+_{max}		
					$\Delta T = 10$ °C	20 °C	30 °C
2	0.8	20	16,079,141	58	1.76	2.3	2.65
3	1	25	15,733,511	80	2.26	2.96	3.39
4	1.1	28	17,945,957	64	2.55	3.26	3.74

The other regions of the computational domain are meshed using tetrahedral elements. The grid cells are concentrated near the walls using ten inflation layers. This is done to properly resolve the turbulent boundary layer and to adequately capture the gradients of the physical properties within.

The grid discretizing the cross-section of the shaft in the x - y plane is displayed in Fig. 5-a, and that around the building is presented in Fig. 5-b. The metrics used to report the quality of the grid are the maximum aspect ratio and the maximum wall y^+ value. Even though it is generally recommended that the maximum aspect ratio be close to 1, Knupp [50] noted that the threshold values for geometry-based metrics lack mathematical-based justification and are selected arbitrarily based on the user's judgment. In this problem, high aspect ratios are inevitable since the turbulent nature of the flow requires one-directional grid refinement. For an adequate boundary layer resolution, it is recommended that the maximum wall y^+ value, which is a solution-dependent metric, be less than 5 [51]. Table 5 presents the meshing parameters employed in addition to the maximum aspect ratio (ζ_{max}) for each geometric case. The maximum wall y^+ value, computed for all cases considered, does not exceed 4.

2.3.1. Grid-independent solutions

Grid-independent solutions are attained by performing computations over successively refined grids for each of the considered geometries for $\Delta T = 10$ °C in Scenario A. Table 6 lists, for each grid, the number of elements, thickness of the first inflation layer, the maximum wall y^+ value, the mass flow rate in the shaft ($\dot{m}$), and the rate of heat emitted by the walls ($\dot{Q}$). The table also lists

Table 6

Comparison of the mesh parameters, mass flow rate ($\dot{m}$), and rate of heat exchange with the shaft wall ($\dot{Q}$) for the four successively refined grids for the case with $\Delta T = 10^\circ\text{C}$.

	$b = 2\text{ m}$				% Relative difference		
	Coarser	Coarse	Medium	Fine	Coarser-Fine	Coarse-Fine	Medium-Fine
N_E	4,967,117	12,411,944	16,079,141	36,355,898	–	–	–
t_1 (mm)	3	2	0.8	0.6	–	–	–
$y_{\max}^+$	6.41	4.96	1.76	1.32	–	–	–
$\dot{m}$ (kg/s)	3.9	3.94	3.76	3.68	5.98	7.07	2.18
$\dot{Q}$ (W)	8290	8073	7556	7340	12.94	9.99	2.94
L_2 norm of local Nu profile (%)	–	–	–	–	14.7	13.13	3.6
	$b = 3\text{ m}$				% Relative difference		
	Coarser	Coarse	Medium	Fine	Coarser-Fine	Coarse-Fine	Medium-Fine
N_E	5,527,075	9,349,292	15,733,511	28,740,698	–	–	–
t_1 (mm)	3	2	1	0.8	–	–	–
$y_{\max}^+$	6.58	4.81	2.26	1.8	–	–	–
$\dot{m}$ (kg/s)	6.79	6.756	6.65	6.68	1.65	1.14	0.45
$\dot{Q}$ (W)	12 262	11 984	11 276	11 204	9.44	6.96	0.64
L_2 norm of local Nu profile (%)	–	–	–	–	11.4	9.78	1.9
	$b = 4\text{ m}$				% Relative difference		
	Coarser	Coarse	Medium	Fine	Coarser-Fine	Coarse-Fine	Medium-Fine
N_E	6,608,236	8,449,794	17,945,957	28,981,550	–	–	–
t_1 (mm)	3	2	1.1	0.9	–	–	–
$y_{\max}^+$	6.59	4.54	2.55	2.05	–	–	–
$\dot{m}$ (kg/s)	9.85	10.09	9.812	9.804	0.47	2.92	0.08
$\dot{Q}$ (W)	15 936	15 710	15 220	14 812	7.59	6.06	2.75
L_2 norm of local Nu profile (%)	–	–	–	–	10.44	8.73	3.64

the percent relative differences in $\dot{m}$ and $\dot{Q}$ between the medium and fine grid solutions, in addition to the L_2 norm of the local Nu profile computed for the coarser, coarse and medium grids with respect to the fine grid. Comparisons of the computed profiles, along the shaft, of the local Nusselt number using the coarser, coarse, medium, and fine grids for the different shaft widths are presented in Fig. 6. These comparisons, in addition to the low values of the percent relative differences and L_2 Norms presented in Table 6, indicate that for all geometries, the medium grid generates results within a reasonable accuracy (less than 4%).

2.4. Comparison with natural convection over a vertical flat plate

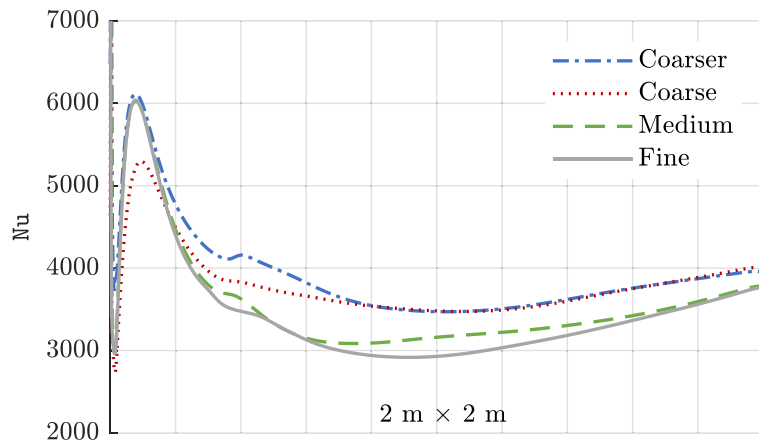
In this section, we first establish that the flow in the vertical shaft is in a developing regime. To represent this, the shaft walls are modeled as four independent vertical flat plates. We then evaluate the validity of the numerical setup and turbulence model by simulating buoyant flow over a single vertical flat plate and comparing the predicted Nusselt numbers with the well-established correlations of Churchill and Chu. Finally, to assess the accuracy of treating the developing shaft flow as four flat-plate flows, we compare the Nusselt numbers obtained from the shaft simulation with those given by the Churchill and Chu correlation.

For free convection over an isothermally heated vertical flat plate, the thermal boundary layer thickness is predicted using the well-established correlation [52,53]:

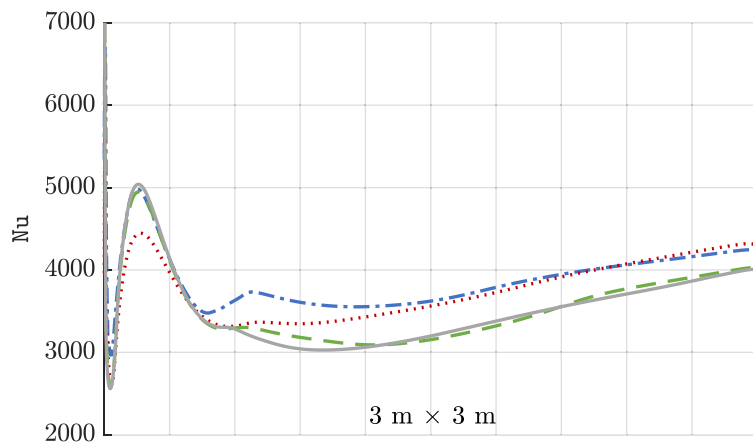
$$\delta_{\text{th}}(z) \approx 0.13 z \text{Ra}_z^{\frac{1}{10}} \text{Pr}^{\frac{1}{12}} \quad (11)$$

where z is the distance from the leading edge of the plate. For the cases considered in this study, the thickness of the boundary layer, computed at the downstream tip of the plate, ranges between 17 cm and 20 cm, which is 10 times smaller than the smallest shaft width considered. This suggests that the flow in the shaft is developing, and as such, the thermal behavior of the flow can be approximated as that over four isolated vertical plates with non-interacting boundary layers.

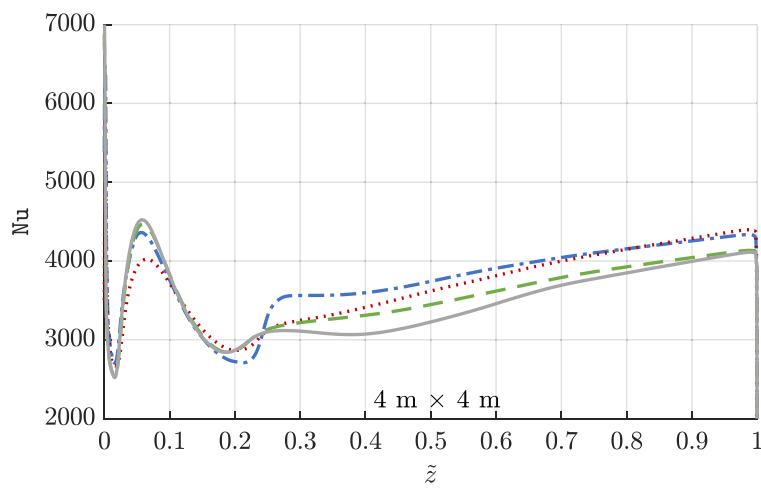
The lack of experimental and numerical studies investigating turbulent natural convection flows in three-dimensional open-ended ducts renders the validation of results a difficult task. To evaluate the validity of the numerical setup and to verify the accuracy of the turbulence model employed in the shaft simulations, we carried out CFD calculations of buoyant flow over a vertical flat plate using the same numerical configuration. The resulting average Nusselt number, computed over the interval $z = 0$ to $z = z^*$, was then compared with the correlation of Churchill and Chu. To this end, we consider two scenarios: one with a plate height of 32 m, and another with a height of 7 m. In both cases, the flat plate, maintained at a temperature 10°C above the ambient temperature, is placed in a large domain whose sides are treated as walls maintained at the ambient temperature. The physical setup is identical to the one employed in our three-dimensional domain used for the shaft, presented in Section 2.2. Fig. 7 shows the percent relative difference between the numerical and empirical average $\overline{\text{Nu}}$ (Eq. (19)), as a function of elevation normalized



(a)



(b)



(c)

Fig. 6. Comparison of the local Nusselt Number (Nu) profile along $\tilde{z}$ computed using the coarser, coarse, medium and fine grids for (a) $b = 2$ m, (b) $b = 3$ m and (c) $b = 4$ m at $\Delta T = 10$ °C.

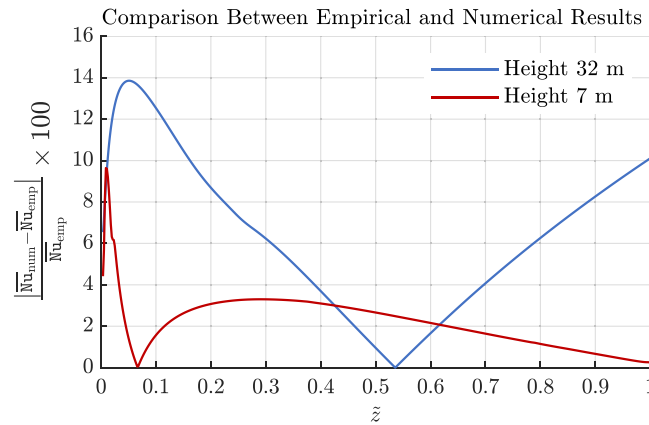


Fig. 7. Comparison of Nusselt number, averaged over distance $\bar{z}$ from leading edge, of buoyant flow over an isothermal flat plate: empirical (Eq. (19)) vs. numerical results.

by the plate height. The results show that, the numerical predictions are within 15% or less of those given by the Churchill and Chu correlation, with the 7 m plate aligning more closely with the empirical correlation than those for the 32 m case. The observed difference may be attributed to the following reasons. First, Eq. (19) accounts for a smooth transition from laminar to turbulent regimes [54]. As such, in the lower part of the plate, we expect the numerical results to depart from the empirical results since the numerical model assumes the flow to be fully turbulent along the entire plate, whereas the empirical formulation accounts for a global laminar–turbulent transition. Second, experimental data that reported average $\overline{Nu}$ correlation for the range of Rayleigh number considered in this study are scarce [54]. Third, intensification of turbulence effects at higher Rayleigh numbers increases the uncertainties associated with turbulence closure using RANS models.

Having established that the flow is still developing and that the numerical setup is reliable, we compare the average Nusselt number over a selected segment of the shaft wall in the three-dimensional model with the empirical correlation of Churchill and Chu. The comparison is performed for the widest shaft (4 m) with the lowest temperature difference ($\Delta T = 10^\circ\text{C}$), since the effects of wall interactions are minimal in this configuration. We also considered 2 cases, where in one case the selected segment extends over $H = 7$ m of the shaft wall, whereas in the other case it extends over the entire wall ($H = 32$ m). For both cases, the average Nusselt number, $\overline{Nu}_H$, is evaluated at the downstream tip (i.e. $z = H$) of the selected segment. The results confirm that the heat transfer between the buoyant flow and the shaft wall exhibits a similar behavior to that over a flat plate of the same area, where the percent relative difference in $\overline{Nu}_H$ between the flat plate (Churchill and Chu) and the shaft (3D CFD) is 3.2% for the 7 m plate and 7.91% for the 32 m plate. Note that, for the 7 m case, the choice of the wall area over which $\overline{Nu}$ is calculated is based on boundary layer development and flow separation. Due to the sharp edge at the shaft inlet, the flow separates, which is not the case for the turbulent natural convection flow along flat plates. Because of that, the area of interest starts beyond flow reattachment. For a shaft size of 4 m and temperature difference of $\Delta T = 10^\circ\text{C}$, flow reattachment occurs at an elevation of around 1.9 m. Accordingly, the value of $\overline{Nu}$ for the 7 m case is computed over an area that extends from 2 m to 9 m.

2.5. Simplified models

Even though three-dimensional numerical simulations can yield accurate results, they are often computationally expensive and less practical for investigating a wide range of scenarios. Simplified or reduced-order models address this limitation by introducing reasonable assumptions that simplify the governing equations, while also offering physical insight into the dominant processes. In doing so, they enable not only a more efficient optimization of design parameters but also the exploration of the limits of the problem under study. Moreover, these models provide a qualitative validation of numerical simulations, rendering them a valuable complement to high-fidelity CFD studies. In addition, simplified models can be informed by CFD results and enhanced to become more accurate. The improved models can then be used in large frameworks where CFD simulations in a large domain containing many buildings and shafts becomes prohibitive. A hybrid approach that combines CFD and calibrated or CFD informed simplified models would be more viable solution. Hence, in this section, we present two simplified models for buoyancy-driven flows in vertical shafts. The first model is a stack model, which only estimates integral quantities such as the mass flow rate and wall heat transfer rate. The second model is a one-dimensional model that additionally takes into account axial variations of the local field variables.

2.5.1. Stack model

The stack model is based on converting the net buoyancy pressure difference between the top (exit) and bottom (inlet) of the shaft, arising from the hydrostatic imbalance, to a dynamic pressure, while including friction and losses at the inlet and outlet of the shaft [55–58]. The model assumes linear variation of the mean air temperature along the shaft, whereas the density variation

follows the Boussinesq approximation of an ideal gas. For the developing flows considered here, heat exchange between the wall and air flow in the shaft is limited by a finite heat transfer coefficient, $\bar{h}$, such that under this weak heating, the air temperature along the shaft stays near the inlet temperature. Under these conditions, the rate of heat exchange at the wall can be approximated as $\dot{Q} \approx \bar{h} \Pi H (T_w - T_i)$ with some effective $\bar{h}$. The buoyancy established air mass flow predicted by the model is

$$\dot{m} = \left(\frac{\rho_i^2 A^2 g H \beta}{c_p (1 + K_{\text{tot}})} \bar{h} \Pi H (T_w - T_i) \right)^{1/3} \quad (12)$$

where ρ_i is the density at the inlet, T_i is the temperature at the inlet, A is the shaft cross-sectional area, Π is the shaft cross section perimeter, $D_h = 4A/\Pi$ is the hydraulic diameter, and K_{tot} is the total loss coefficient accounting for losses at inlet, outlet, and due to friction, i.e. $K_{\text{tot}} = K_i + K_o + \bar{f} H/D_h$, $\bar{f}$ is the Darcy friction factor, and $\bar{h}$ is the average convection heat transfer coefficient.

Note that solving Eq. (12) requires an iterative solution procedure, since $\bar{f}$ depends on Reynolds number, which is not known beforehand. Moreover, for the ideal case, K_{tot} is set to 0 in Eq. (12).

2.5.2. One-dimensional model

The one-dimensional model takes into account the variation of the local flow variables, taken to be uniform over the shaft cross section, in the axial direction, z . The momentum equation is approximated as a balance between the net buoyancy pressure and the friction losses over an infinitesimal control volume extending from z to $z + dz$. The inlet and outlet static pressures are related to the ambient conditions at the corresponding elevations by taking into account the dynamic head and inlet/outlet losses. The energy equation is modeled as balance between the rate of heat transfer at the wall and the increase in the flow enthalpy. The simplified governing equations are presented in Eqs. (13)–(17). A detailed derivation of these equations is provided in [Appendix C.2](#)

$$\dot{m} = \rho(z)u(z)A = \text{cst} \quad (13)$$

$$\int_0^H (\rho_\infty - \rho(z))gAdz = \int_0^H \frac{f\rho u^2}{8}\Pi dz + \Delta p_{\text{loss}}A \quad (14)$$

$$\Delta p_{\text{loss}} = K_i \frac{1}{2} \rho_i u_i^2 + K_o \frac{1}{2} \rho_o u_o^2 \quad (15)$$

$$\frac{dT}{dz} = \frac{q^- \Pi}{\dot{m} c_p} \quad (16)$$

$$\rho(z) = \frac{p_i}{RT(z)} \quad (17)$$

where u is the axial velocity, and the subscripts i , o , and ∞ refer to inlet, outlet and ambient respectively. The heat transfer rate per unit area of the shaft wall is

$$q^- = \bar{h}(T_w - T(z))$$

The solution procedure is outlined in Algorithm 1 in [Appendix C.2](#).

Since the flow is developing and the boundary layer thickness is small compared to the shaft size, the heat transfer coefficient ($\bar{h}$) at the walls is approximated, for both models, based on the Churchill and Chu Nusselt number correlation for the vertical plate [54] presented in Eq. (19), so that

$$\bar{h} = \frac{\overline{\text{Nu}} \lambda}{H} \quad (18)$$

and

$$\overline{\text{Nu}} = \left\{ 0.825 + \frac{0.387 \text{Ra}^{1/6}}{[1 + (0.492/\text{Pr})^{9/16}]^{8/27}} \right\}^2 \quad (19)$$

valid over a wide range of Ra and Pr numbers. The Rayleigh number is

$$\text{Ra} = \frac{g\beta(T_w - T_\infty)H^3}{\nu\alpha}, \quad (20)$$

H (m) is the height of the shaft, where λ is the thermal conductivity of air, ν (m^2/s) is the kinematic viscosity, and α (m^2/s) is the thermal diffusivity. The fluid properties are evaluated at the film temperature ($T_f = (T_w + T_\infty)/2$).

For both models, the friction factor, $\bar{f}$, is approximated using the Swamee Jain correlation [59] for pipe flows using a hydraulic diameter equal to the shaft width. We next compare the air mass flow rates predicted by the simplified models with those obtained from detailed CFD simulations. The observed discrepancies are primarily attributed to the three-dimensional, developing nature of buoyant flow in the shaft. Following the comparison, we examine key features of this flow in greater detail.

3. Results

For each of the considered shaft widths ($b = 2$ m, 3 m, and 4 m), the flow-field of the thermally induced buoyancy-driven flow in the building vertical shaft is studied for three values of the wall temperature ($T_w = 35$ °C, 45 °C, and 55 °C). This section first compares results the three-dimensional simulations for Scenarios A and B, and then compares the results of Scenario A with

Table 7

Comparison between $\dot{m}$ and $\dot{Q}$ predicted by the three-dimensional simulations when the far-field planes are modeled as stationary walls versus when they are treated as pressure outlets.

b (m)	Scenario A - $\dot{m}$ (kg/s)			Scenario B - $\dot{m}$ (kg/s)			% Relative difference		
	ΔT (°C) = 10	20	30	10	20	30	10	20	30
2	3.76	5.18	6.18	3.81	5.15	6.08	1.5	0.55	1.6
3	6.65	9.22	11.10	6.82	9.16	10.93	2.60	0.56	1.61
4	9.81	13.71	16.63	10.61	14.12	16.55	8.13	2.96	0.47
b (m)	Scenario A - $\dot{Q}$ (kW)			Scenario B - $\dot{Q}$ (kW)			% Relative difference		
	ΔT (°C) = 10	20	30	10	20	30	10	20	30
2	7.56	19.80	34.44	7.58	19.59	34.26	0.20	1.10	0.52
3	11.28	29.82	52.59	11.22	29.56	51.84	0.56	0.87	1.45
4	15.22	39.82	70.15	15.14	39.88	69.70	0.50	0.16	0.65

the predictions of the simplified models introduced in Section 2.5. The subsequent analysis focuses on Scenario A, reporting its ventilation effectiveness. Then, we explore the three-dimensional nature of the flow by inspecting the streamlines at the inlet to the shaft, axial velocity profiles, Nusselt Number variations along the shaft, in addition to isotherms in the shaft cross-section at various elevations. Finally, we propose correlations for the average Nusselt number, the average friction factor, reattachment length and core flow length in terms of Rayleigh number and the shaft aspect ratio.

3.1. Mass flow rate and wall heat flux

3.1.1. Impact of boundary conditions on the mass flow rate and wall heat transfer rate

Table 7 displays the mass flow rate and wall heat transfer rate computed by the three-dimensional simulations of Scenario A (far-field walls) and Scenario B (far-field outlets), together with their relative differences. Both scenarios exhibit the same qualitative behavior, where $\dot{m}$ and $\dot{Q}$ increase with b and ΔT . We observe that the relative difference in $\dot{m}$ between Scenarios A and B is largest for the widest shaft maintained at the lowest ΔT ($\approx 8.13\%$), where the difference generally decreases with an increase in ΔT . This trend indicates that stronger buoyancy forces, associated with larger ΔT , reduce the sensitivity of the solution to the far-field boundary conditions. Meanwhile, for a fixed ΔT , wider shafts exhibit larger discrepancies between the two cases, suggesting that bringing the shaft walls closer to the far-field boundaries intensifies the influence of the artificial flow suppression imposed by the stationary wall condition in Scenario A. Regarding the wall heat transfer rate, the two cases remain in close agreement, with relative differences below 1.5%. The observed trends show that mass flow rates and the wall heat transfer rates are robust to the far-field treatment when the domain is sufficiently large. Due to this close agreement in the results, the discussion in the following sections is limited to Scenario A.

3.1.2. Comparison with simplified models: Scenario A

In Table 8, we present the mass flow rates predicted by the stack model and the one-dimensional model, along with the percent relative difference with the mass flow rate predicted by the three-dimensional numerical simulations of Scenario A. We observe that both models produce qualitatively similar results and follow the trends observed in the three-dimensional simulations, with the stack model outperforming the one-dimensional model.

Deviations of the stack model predictions from the numerical results of Scenario A are larger for wider shafts maintained at lower temperatures. These discrepancies arise mainly from two sources: the average convection heat transfer coefficient, $\bar{h}$, and the average friction factor, $\bar{f}$. As argued earlier, $\bar{h}$ estimated from the Churchill and Chu's Nusselt number correlation provides a reasonable approximation of the thermal behavior in the shaft, which suggests that it is unlikely to be the dominant source of error. As for the friction factor, due to the scarcity of studies that reported the friction factor of buoyancy-driven flows, we resorted to empirical correlations developed for pressure-driven flows. This approximation introduces errors since the empirical correlation employed was established for a fully-developed flow whose boundary layer thickness is comparable to the shaft size. Meanwhile, the buoyancy-driven flow in this study is still developing, and its boundary layer remains less than one-tenth the shaft width. The thinner boundary layer leads to steeper velocity gradients and stronger shearing near the walls, resulting in larger friction factors.

Table 8 shows that the one-dimensional model shows improved agreement with the numerical simulations for smaller shafts maintained at higher temperatures. The observed relative percent differences are attributed to the approximations used for the local values of the convection heat transfer coefficient, h , and the friction factor, f . Although the one-dimensional model attempts to capture local variations in flow variables, there are no empirical data for the local distributions of h and f in free convection flows. To overcome this limitation, h and f are assumed to be uniform along the shaft, where $\bar{h}$ is obtained from Churchill and Chu's Nusselt number correlation and $\bar{f}$ is computed from the empirical correlations established for pressure-driven flows.

In what follows, we show that the stack model and one-dimensional model predictions can be improved by using an average friction factor, $\bar{f}$, calculated according to

$$\bar{f} = \frac{8\tau_w}{\rho u^2} = \frac{8\tau_w \rho A^2}{\dot{m}^2} \quad (21)$$

Table 8

Mass flow rate predicted by the stack model and the one-dimensional model and the percent relative difference with respect to the numerical simulations.

b (m)	$\dot{m}_{\text{stack, losses}}$ (kg/s)			% Relative difference scenario A		
	ΔT (°C) = 10	20	30	10	20	30
2	3.93	5.35	6.41	4.52	3.28	3.72
3	7.83	10.65	12.76	17.74	15.51	14.95
4	12.72	17.32	20.73	29.66	26.33	24.65
b (m)	$\dot{m}_{\text{1D, losses}}$ (kg/s)			% Relative difference scenario A		
	ΔT (°C) = 10	20	30	10	20	30
2	4.49	6.05	7.22	19.41	16.8	16.83
3	8.97	12.1	14.44	34.89	31.24	30.09
4	14.63	19.71	23.42	49.13	43.76	40.83

Table 9

Wall shear stress and average friction factor from numerical results and corresponding mass flow rate predicted by the simplified models with the percent relative difference with respect to the numerical results.

b (m)	Wall shear stress (Pa)			$\bar{f}$		
	ΔT (°C) = 10	20	30	10	20	30
2	0.0051	0.0089	0.0123	0.054	0.049	0.046
3	0.0048	0.0084	0.0117	0.082	0.073	0.069
4	0.0046	0.0081	0.0112	0.114	0.101	0.094
b (m)	$\dot{m}'_{\text{stack, losses}}$ (kg/s)			% Relative difference - Scenario A		
	ΔT (°C) = 10	20	30	10	20	30
2	3.69	5.06	6.08	1.86	2.32	1.62
3	7.25	9.95	11.96	9.02	7.92	7.75
4	11.67	16.02	19.29	18.96	16.85	16.00
b (m)	$\dot{m}'_{\text{1D, losses}}$ (kg/s)			% Relative difference - Scenario A		
	ΔT (°C) = 10	20	30	10	20	30
2	3.71	5.08	6.05	1.33	1.93	2.1
3	7.22	9.95	11.91	8.57	7.92	7.3
4	11.51	16.00	19.32	17.33	16.7	16.18

where the average wall shear stress is obtained from the numerical simulations, and the density ρ is taken to be the film density.

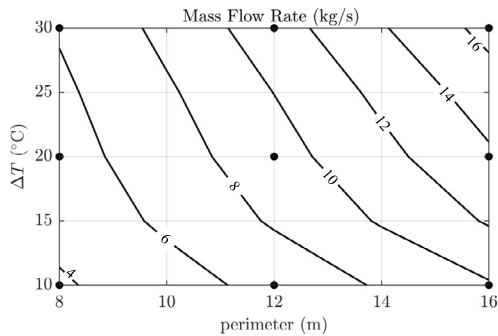
Table 9 lists the average friction factor, $\bar{f}$, the corresponding mass flow rate computed using the simplified model, and the percent relative difference with respect to the three-dimensional simulations. For a given shaft size, the table shows that $\bar{f}$ decreases as ΔT is increased. For larger ΔT , stronger buoyancy results in a larger shear at the wall as well as larger mass flow rate. For $b = 4$ m, τ_w increases from 0.0046 Pa to 0.011 Pa as ΔT is increased from 10 °C to 30 °C. The mass flow rate correspondingly increased from 9.81 kg/s to 16.6 kg/s. It follows from Eq. (21) that $f(\Delta T = 30 \text{ °C})/f(\Delta T = 10 \text{ °C})$ scales as 0.835, which matches the corresponding ratio of 0.824 obtained from the table. The table also shows that the friction factor increases for larger shafts at a given ΔT . The flat plate model approximation to the developing buoyant flow in the shaft predicts that the average shear is constant for constant ΔT and plate length H . Thus, as the shaft width increases for a given ΔT , τ_w does not change (confirmed by the τ_w values predicted by CFD), so that f depends largely on u^{-2} or $A^2/\dot{m}^2$. For $\Delta T = 30 \text{ °C}$, $A^2/\dot{m}^2$ increases by a factor of 2.2 as b increases from 2 to 4 m, which is sufficiently close to the corresponding ratio of 2.04 obtained from the table. To further demonstrate that using Eq. (C.33) to compute $\bar{f}$ is an important source of discrepancy, we show in Table 9 that both models predict the mass flow rate more accurately when using the $\bar{f}$ from the numerical simulations, with the one-dimensional model slightly outperforming the stack model. A comparison between the results of Tables 8 and 9 shows that the one-dimensional model is more sensitive to the friction factor than the stack model. Nevertheless, because of its more general formulation, it is preferable in applications where the stack model cannot be applied, such as shafts with variable cross-sectional areas. Despite the observed improvement with $\bar{f}$, both models still fail to predict the shaft mass flow rate with acceptable accuracy—especially for wider shafts at low temperature differences. The shortfall arises because the model does not resolve the cross-sectional flow field and therefore misses three-dimensional effects, including corner thermal/dynamic behavior and entry recirculation. In particular, the sharp-edge inlet/outlet losses are evaluated based on a pressure-driven flow, with a velocity profile that differs markedly from the buoyancy-driven profile shown in Fig. 10.

3.1.3. Ventilation effectiveness: Scenario A

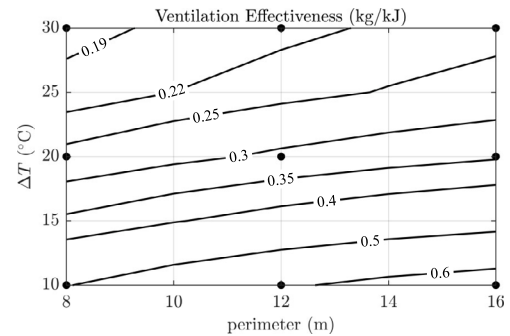
The capacity of the heated shaft to ventilate the surrounding area, at pedestrian level, is measured by the mass flow rate ($\dot{m}$) established by the buoyant flow in the shaft, listed in Table 7 for the different cases considered. Table 10 lists the mass flow rate per unit heat transfer rate, which serves as a measure of the ventilation effectiveness ($\dot{m}/\dot{Q}$). Although numerous studies have investigated buoyancy-driven flows in vertical channels, most have been restricted to two-dimensional configurations

Table 10
Predicted values of the mass flow rate per unit heat transfer rate.

b (m)	$\dot{m}/\dot{Q}$ (kg/kJ)		
	10	20	30
2	0.497	0.262	0.179
3	0.590	0.309	0.211
4	0.645	0.344	0.237



(a) Mass flow rate as a function of shaft width and temperature difference. Perimeter = $4b$.



(b) Ventilation Effectiveness as a function of shaft width and temperature difference. Perimeter = $4b$.

Fig. 8. Contour plots of the mass flow and ventilation effectiveness.

and have focused on indoor air quality enhancement or cooling of electronic equipment. The present analysis evaluates the ventilation effectiveness of three-dimensional shafts designed for outdoor environments, thereby extending the applicability of natural convection to outdoor building-scale ventilation scenarios.

In Table 7, we observe that increasing the temperature difference yields a higher mass flow rate, which can be attributed to the intensification of buoyancy forces and the corresponding rise in bulk velocity. As for the wall heat transfer rate, it is directly proportional to the convection heat transfer coefficient and the area through which heat is transferred. We have previously shown that the thermal behavior of the shaft is similar to that of four separate vertical plates. Hence, we expect the convection heat transfer coefficient for this developing flow to be limited to the temperature difference rather than the shaft size. For a fixed temperature difference, the increase in the total heat transfer rate in wider shafts is attributed to the increase in the heat transfer area. Meanwhile, for a fixed shaft size, the increase in $\dot{Q}$ in hotter shafts is attributed to the intensification of turbulence effects, which promotes heat transfer and increases the convection heat transfer coefficient.

Contours of the established mass flow rate versus the temperature difference and the shaft size are plotted in Fig. 8(a). The dependence of the ventilation effectiveness on the shaft size b and the temperature difference ΔT is depicted in the contours plot of Fig. 8(b), where two trends can be observed. In the first trend, the ventilation effectiveness increases as the temperature difference decreases. Keeping the shaft size constant and increasing the wall temperature, intensifies the buoyancy forces, resulting not only in stronger inertial effects but also in larger velocity gradients and shearing near the walls. The stronger buoyancy forces increase the mass flow rate in the shaft, however, shearing counteracts flow acceleration and increases friction losses. Moreover, the geometric confinement amplifies mixing under stronger buoyancy conditions (i.e. higher ΔT), thereby enhancing convective heat transfer. Hence, as ΔT increases, $\dot{Q}$ increases more rapidly than $\dot{m}$, which results in a decrease in the ventilation effectiveness of the shaft. In the second trend, the ventilation effectiveness increases as the shaft size increases. Enlarging the shaft while maintaining its temperature fixed, increases the flow capacity. Moreover, the weaker geometric confinement also reduces mixing intensity, where the growth in total wall heat transfer is then driven mainly by the larger heat-transfer area rather than a strong rise in local heat-transfer coefficients. As such, as the shaft size increases, the increase in $\dot{Q}$ is slower than that in $\dot{m}$, yielding a higher ventilation effectiveness in the shaft. Contour plots of $\dot{Q}$ are shown in Fig. B.1 in Appendix B.

3.2. Streamlines and flow separation

The streamlines at the inlet section of the $b = 3$ m shaft, presented in Figs. 9(a)–9(c) for the three shaft wall temperatures considered, show that flow separation occurs at all values of Ra . Similar behavior has been reported in [6,14] where the flow recirculation was associated with the relatively high value of Gr (Ra) number. At higher Gr (Ra) values, air enters the shaft with higher velocity magnitudes as a consequence of stronger buoyancy forces. Moreover, at the inlet to the shaft, the flow cross-section experiences a sudden contraction, resulting in a higher bulk velocity and creation of a region of adverse pressure gradient near the walls that causes flow separation. The high pressure gradients normal to the streamlines as the high inertia flow makes a sharp turn

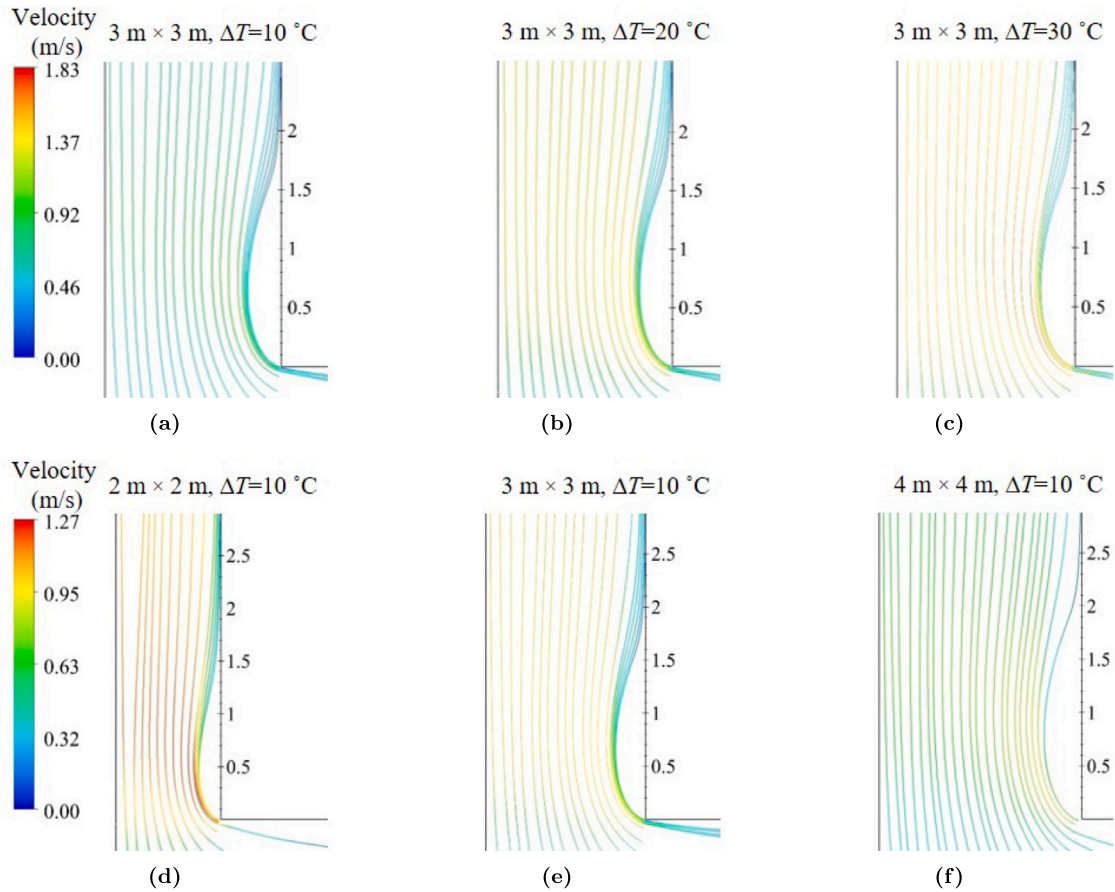
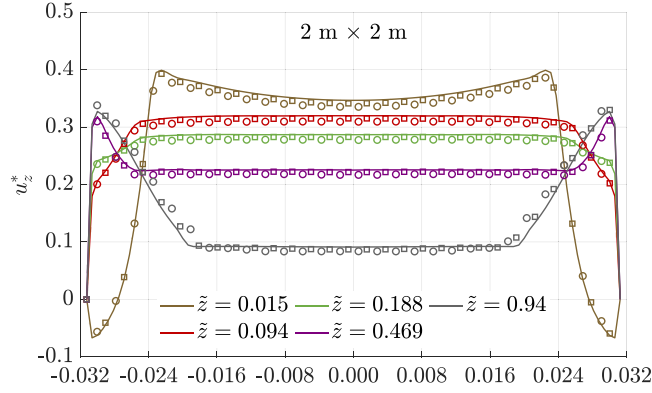


Fig. 9. (a–c) Impact of the temperature difference on flow separation for the shaft with $b = 3$ m, (d–f) Impact of the shaft width on flow separation for a temperature difference of 10 °C.

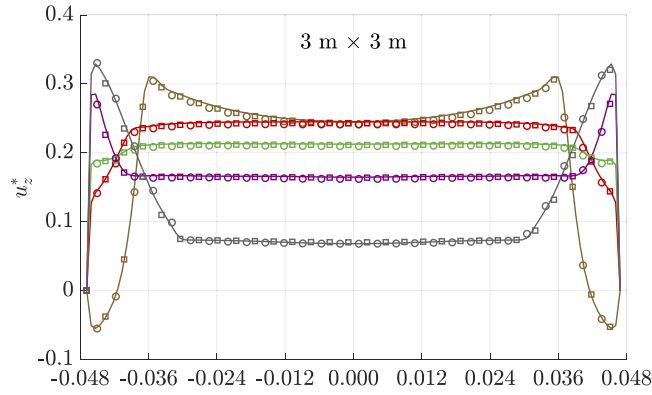
are unsustainable for a small radius of curvature, thus necessitating the flow to separate at the inlet corner to reduce the curvature and, in effect, create a recirculation zone. The streamlines presented in Figs. 9(a)–9(c) demonstrate that the impact of Ra (or ΔT) on the reattachment length is insignificant for a given shaft width. Flow reattachment occurs at the elevation where the buoyancy forces overcome the pressure gradient effects. For a given shaft width, increasing Ra has opposing effects. On the one hand, the velocity increases yielding stronger pressure gradient effects, which delays the flow reattachment. On the other hand, the buoyancy force, which opposes the pressure gradient, increases. So for a given shaft width, the increase in the buoyancy force that results from increasing Ra , balances the increase in the pressure gradient rendering the reattachment length unaffected. Increasing the shaft width, while holding constant the temperature difference between shaft wall and the ambient, causes a delay in the reattachment of the separated flow. As shown in the streamlines presented in Figs. 9(d)–9(f) for $\Delta T = 10$ °C, flow separation occurs at the minimum Ra for all shaft widths, which implies that flow separation occurs for all cases considered in this study. Moreover, it is evident that in larger shafts (compare plots in Figs. 9(d)–9(f)), flow reattachment is delayed despite the decrease in the average bulk velocity. The increase in the shaft width results in weaker viscous forces when compared to inertia, implying that the impact of the larger shaft width outweighs that of the reduction in the average bulk velocity. At a fixed temperature difference, the boundary layer for all shaft widths is subjected to the same conditions, hence, the buoyancy force remains unaffected as the shaft width increases, which suggests the presence of larger pressure gradients in the recirculation region to compensate for the weaker viscous forces. Therefore, in larger shafts, the delay in flow reattachment is a consequence of the unaffected buoyancy forces and the stronger pressure gradient effects.

3.3. Axial velocity profiles

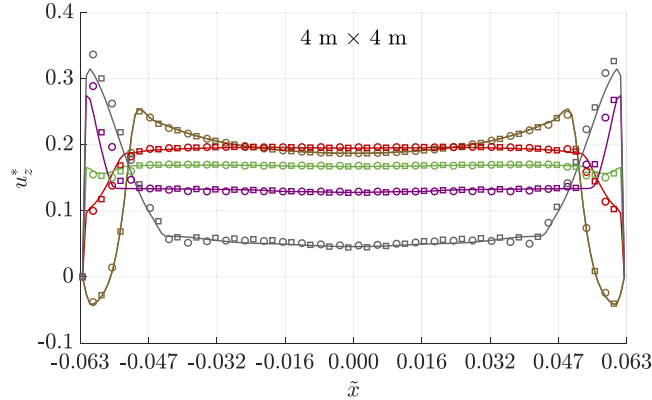
Fig. 10 presents central profiles, along the axis of symmetry, of the axial velocity (non-dimensional), $u_z^*(\bar{x}, 0, \bar{z}^*)$ (or due to symmetry $u_z^*(0, \bar{y}, \bar{z}^*)$), at five horizontal cross sections corresponding to $\bar{z}^* = 0.015, 0.094, 0.188, 0.496$ and 0.94 in the for all studied cases of Scenario A. It can be observed from the figure that, for a given shaft width, $u_z^*(\bar{x}, 0, \bar{z}^*)$ is nearly independent of the



(a)



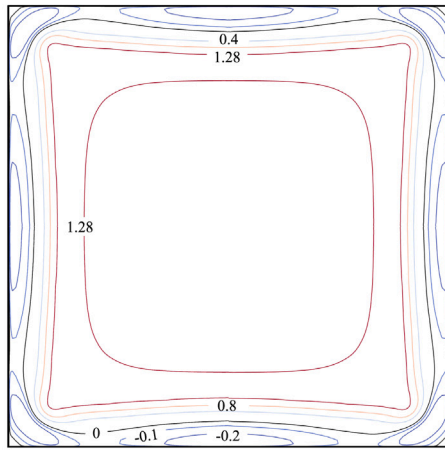
(b)



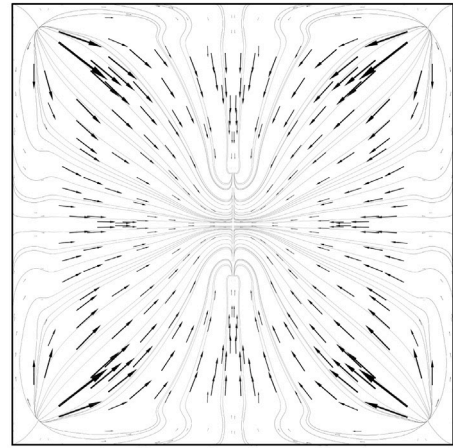
(c)

Fig. 10. Central profiles ($y = 0$) of the axial velocity ($\bar{u}_z^*$) at various shaft cross sections (in the $\bar{x} - \bar{y}$ plane) along the flow direction ($\bar{z}$) for (a) $b = 2$ m, (b) $b = 3$ m, and (c) $b = 4$ m. Solid lines correspond to $\Delta T = 10$ °C, square symbols for $\Delta T = 20$ °C, and circle symbols for $\Delta T = 30$ °C.

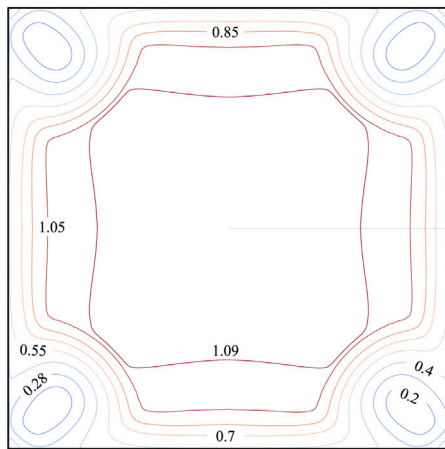
temperature difference. This is attributed to the scaling employed in the normalization procedure of the momentum equation where the buoyancy force, $(\rho_\infty - \rho)\vec{g}$, scales as $(T_w - T_\infty)\vec{g}$ and the inertia term, $\rho(\vec{u} \cdot \nabla \vec{u})$, scales as $\rho_\infty |\vec{u}|^2 / b$. Inspection of the figures reveals that the evolution of the central profiles of u_z^* along the flow direction is similar for the different shaft widths, where the following three zones can be observed: (i) flow separation zone, (ii) core flow zone, and (iii) boundary dominated zone. The axial velocity in the vicinity of the walls at $\bar{z}^* = 0.015$ is negative due to flow reversal in the recirculation zone induced by flow separation. The



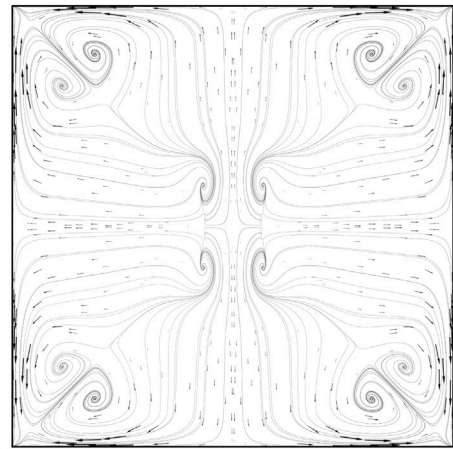
(a1)



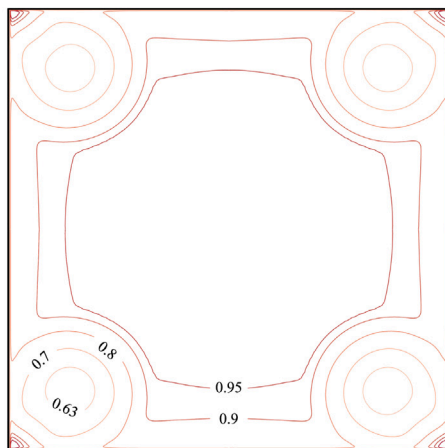
(b1)



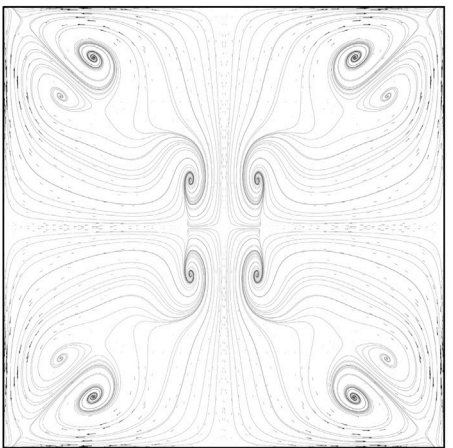
(a2)



(b2)



(a3)



(b3)

Fig. 11a. (a1 - a3) contours of u_z and (b1 - b3) streamlines and vector field in the $x-y$ planes at $z = 0.5, 3$ and 6 m. Case: $b = 3$ m and $\Delta T = 20$ °C.

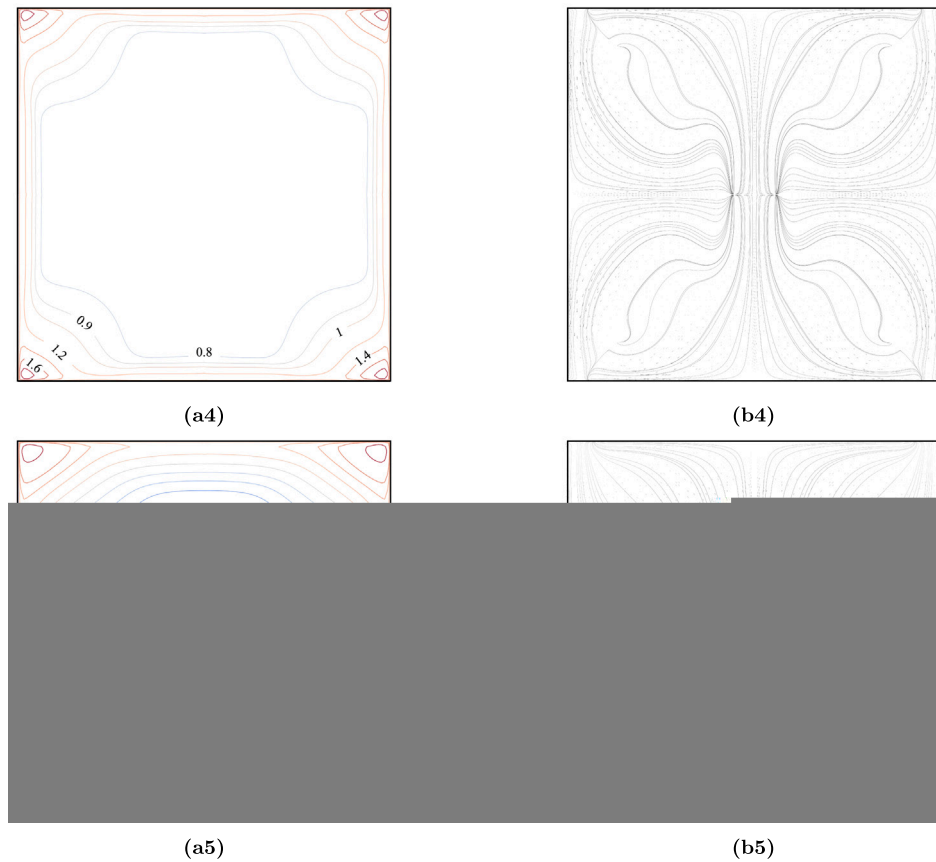


Fig. 11b. (Continued) (a4 - a5) contours of u_z and (b4 - b5) streamlines and vector field in the $x - y$ planes at $z = 15$ and 30 m. Case: $b = 3$ m and $\Delta T = 20$ °C.

recirculation zone reduces the effective area through which air enters the shaft. At $\bar{z}^* = 0.094$, the axial velocity increases away from the walls, reaching its maximum value in the central region of the shaft. In this core flow zone, the buoyancy forces overcome the pressure gradient effects, and consequently, air is pulled upwards. The higher viscous forces near the edges yield smaller axial velocities close to the walls than in the core region of the shaft. At $\bar{z}^* = 0.188$, smaller variations in the axial velocity are obtained. At this elevation, air near the edges starts to accelerate due to stronger buoyancy forces, indicating flow transition into the boundary dominated zone. For a given shaft width, mass conservation dictates that an increase in the axial velocity in one region of the shaft should be accompanied with a decrease in another region. The continuous heating of air near the walls results in flow acceleration in that region, which is accompanied by flow deceleration in the core region of the shaft, as can be observed at $\bar{z}^* = 0.469$ and $\bar{z}^* = 0.938$. In the boundary dominated zone, we observe in Fig. 10 that the maximum velocity is attained at a farther distance from the walls at higher elevations. This is pertinent to the boundary layer development where viscous effects diffuse further towards the centerline at higher elevations.

Additional understanding of the flow field is possible by inspecting contours of the axial velocity component at different $x - y$ sections along the shaft (z -) axis, presented in Fig. 11(a1–a5) for the case $b = 3$ m, $\Delta T = 20$ °C. The streamlines and vector fields in these $x - y$ planes are presented in Fig. 11(b1–b5) at sections $z^* = 0.5, 3, 6, 15$ and 30 m, corresponding respectively to $\bar{z}^* = z^*/z_{max} = 0.015, 0.094, 0.188, 0.469$, and 0.94 . At $z^* = 0.5$ m, the contour of $u_z = -0.2$ (m/s) indicates a re-circulation in the separation region. The impact of the solid boundary at the corner is observed in terms of lower axial velocities due to viscous effects slowing down the flow. This, combined with the constraints imposed by the continuity equation and the symmetry yields the observed streamlines in the corresponding $x - y$ planar sections.

3.4. Centerline velocity

Fig. 12 presents the profiles of $u_z^*(0, 0, \bar{z})$, the dimensionless velocity as a function of $\bar{z} = z/H$ along the centerline of the domain, for all the cases considered in this study. The figure shows that for a given shaft width, the profiles of $u_z^*(0, 0, \bar{z})$ are nearly independent of the temperature difference, ΔT . Fig. 12 also shows that, for all cases, the centerline velocity increases until a local maximum is reached beyond which it decreases monotonously. At the entrance, the increase is dictated by conservation of mass in the presence

of recirculation near the walls, as discussed earlier. After flow reattachment, the decrease is attributed to the greater axial velocities near the walls. As can be seen in Fig. 12, the profiles do not reach a plateau for any of the cases addressed. Hence, a fully developed flow is not reached for any of these cases, which justifies the previous discussion on the flat-plate comparison in Section 2.4.

3.5. Temperature contours

To observe the evolution of the temperature field along the shaft height, isotherms over one fourth of the shaft horizontal cross-sectional area are presented in Fig. 13 at five elevations corresponding to $\bar{z}^* = 0.015, 0.094, 0.188, 0.496$ and 0.94 . The contours are reported for the different shaft widths at a temperature difference of 10°C , and the values are presented in dimensionless form according to T^* in Table 2. The distribution of the isotherms in the shaft is affected by the prevalent mode of heat transfer (conduction vs. convection), which is dictated by the dominance of viscous effects over the buoyancy effects, or vice versa. At $\bar{z}^* = 0.015$, the intensification of the recirculation zone near the corners results in a clustering of the isotherms at this elevation, which belong to zone I. At $\bar{z}^* = 0.094, 0.188$ and 0.496 , the temperature contours in zone II exhibit an opposite behavior to that at $\bar{z}^* = 0.015$, where the isotherms become more clustered near the walls and more spread out around the corners. This is associated with the relatively small axial velocities in the boundary layer. At these elevations, the development of the velocity field near the walls is still in its early stages, and hence convective effects are weak. The closer proximity to the two heated walls, in addition to the slow movement of the flow in the vicinity of the corner yields a stronger conduction heat transfer mode and a more uniform distribution of isotherms in that region. At $\bar{z}^* = 0.94$ in zone III, buoyancy forces dominate the flow leading to enhancement in convective heat transfer due to flow acceleration. Hence, all shafts exhibit a more uniform temperature distribution at higher elevations. However, stronger geometric confinement in smaller shafts intensifies mixing effects and promotes conduction, which results in a more uniform temperature distribution and a higher average temperature when compared to wider shafts.

3.6. Nusselt number and the friction factor

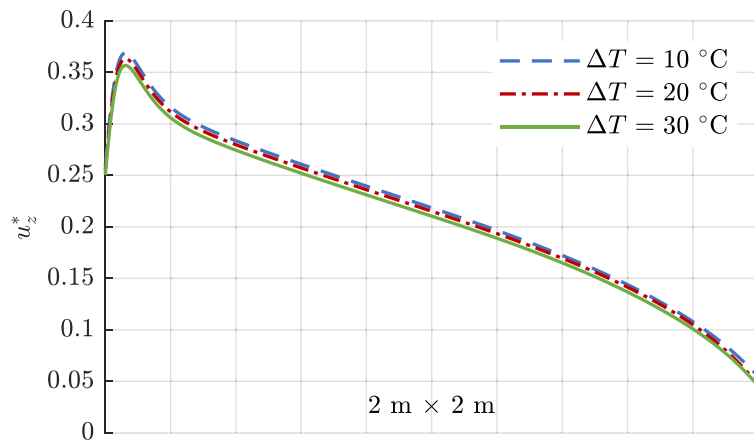
Variations along the shaft wall of the local Nusselt number, Nu , defined in Table 2, are plotted in Fig. 14 as a function of the dimensionless elevation, $\bar{z}$, for all cases considered in this study. Inspection of the subfigures of Fig. 14 reveals that, for a given shaft width, Nu increases with an increase in ΔT . Increasing ΔT strengthens the buoyancy force leading to higher velocities near the walls. This results in a higher convection heat transfer coefficient and consequently a higher value of Nu .

Qualitatively, the variation of Nusselt number along the channel, as the flow goes through the three zones discussed earlier, exhibits a similar trend for all the cases considered. The two key players shaping the Nu profile are the viscous forces and the buoyancy forces. On the one hand, viscous effects near the walls retard the flow and give dominance to heat transfer by conduction, thus decreasing Nu . On the other hand, as air accelerates near the walls due to stronger buoyancy effects, heat transfer by convection is enhanced, and consequently Nu increases. In natural convection flows, an increase in the fluid temperature is coupled with an increase in the flow velocity. As such, the Nu profiles provide insight into the interplay between the viscous and the buoyancy forces. For all the cases considered, Nu increases sharply in the flow-separation zone (denoted by I in Fig. 14), where recirculation occurs and cold air descends near the walls. Recirculation near the walls improves mixing, thus increasing heat transfer by convection. The elevation at which the flow is reattached marks the onset of the core flow zone (denoted by II in Fig. 14), which is characterized by a sharply decreasing Nu . In the core flow zone and the boundary dominated zone (denoted by III in Fig. 14), air near the walls accelerates, and its temperature increases as a result of heat transfer from the walls. In the core flow zone, the sharp decrease in Nu is attributed to the small axial flow velocity near the walls, as depicted by the axial velocity profiles at $\bar{z} = 0.094$ in Fig. 10. The increase in the air thermal resistance as a result of its higher temperature in addition to the small axial velocities near the hot walls explains the sharp decrease in the Nu profile in zone II, where viscous effects are dominant. In the boundary dominated zone (zone III), the local Nu number generally increases downstream the shaft. In this zone, buoyancy forces are dominant and the maximum axial velocity is close to the hot walls as shown in Fig. 10 at $\bar{z}^* = 0.469$ and $\bar{z}^* = 0.94$. The increase in the axial velocity of air near the walls strengthens convection heat transfer effects.

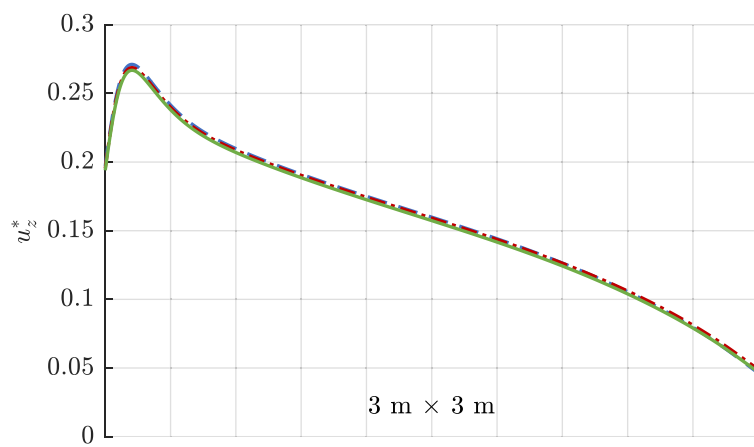
Based on our numerical results, we express in Eqs. (22)–(25) the dimensionless form of the flow reattachment length (ℓ_1^*), the core flow length (ℓ_2^*), the average Nusselt Number ($\bar{Nu}$) and the average friction factor ($\bar{f}$), as functions of Rayleigh number (Ra) and the shaft aspect ratio b/H . The reattachment length ℓ_1 is the axial position where the Nusselt number profile attains its local maximum, while the core flow length ℓ_2 corresponds to the lowest point — beyond reattachment — where the maximum axial velocity occurs near the shaft walls. The coefficient of determination, R value, of the proposed correlations exceeds 0.98. Eqs. (24) and (25) are illustrated in Fig. 15, where the dashed line represents the average Nusselt number and the solid lines represent the friction factor computed for each shaft size. We recall that Rayleigh number depends on the temperature difference and the shaft height, chosen to be the characteristic length scale for the developing flow. At a given temperature difference (or Ra), the Nu range of the associated bar shown in the figure corresponds to the different shaft sizes; $b = 2, 3, 4$ m. The widest range occurs at the highest Rayleigh number considered, where the smallest $\bar{Nu}$ value corresponds to the smallest shaft. As argued in Section 3.5, the geometric flow confinement in the shaft of smaller width promotes conduction heat transfer, especially at higher elevations, which results in a smaller $\bar{Nu}$ when compared to shafts of larger width.

$$\ell_1^* = 0.6504 \left(\frac{b}{H} \right)^{0.6217} (Ra)^{-0.03484} \quad (22)$$

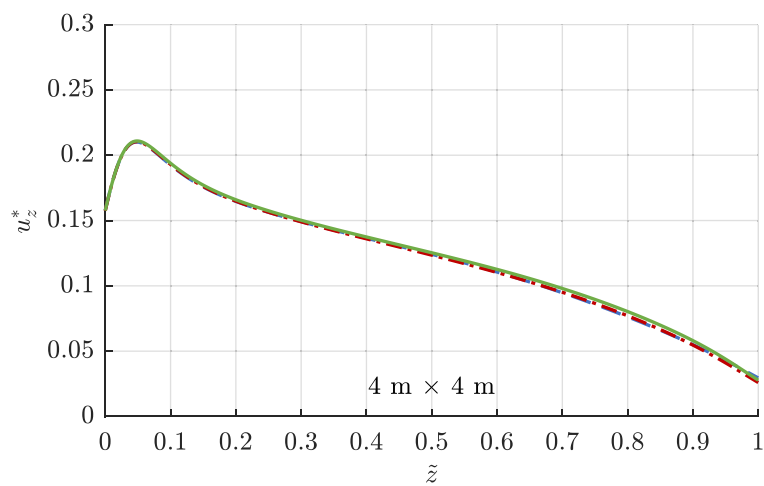
$$\ell_2^* = 0.0385 \left(\frac{b}{H} \right)^{-0.478} (Ra)^{-0.02082} \quad (23)$$



(a)



(b)



(c)

Fig. 12. Centerline dimensionless velocity profiles along $\tilde{z}$ for (a) $b = 2$ m, (b) $b = 3$ m, and (c) $b = 4$ m.

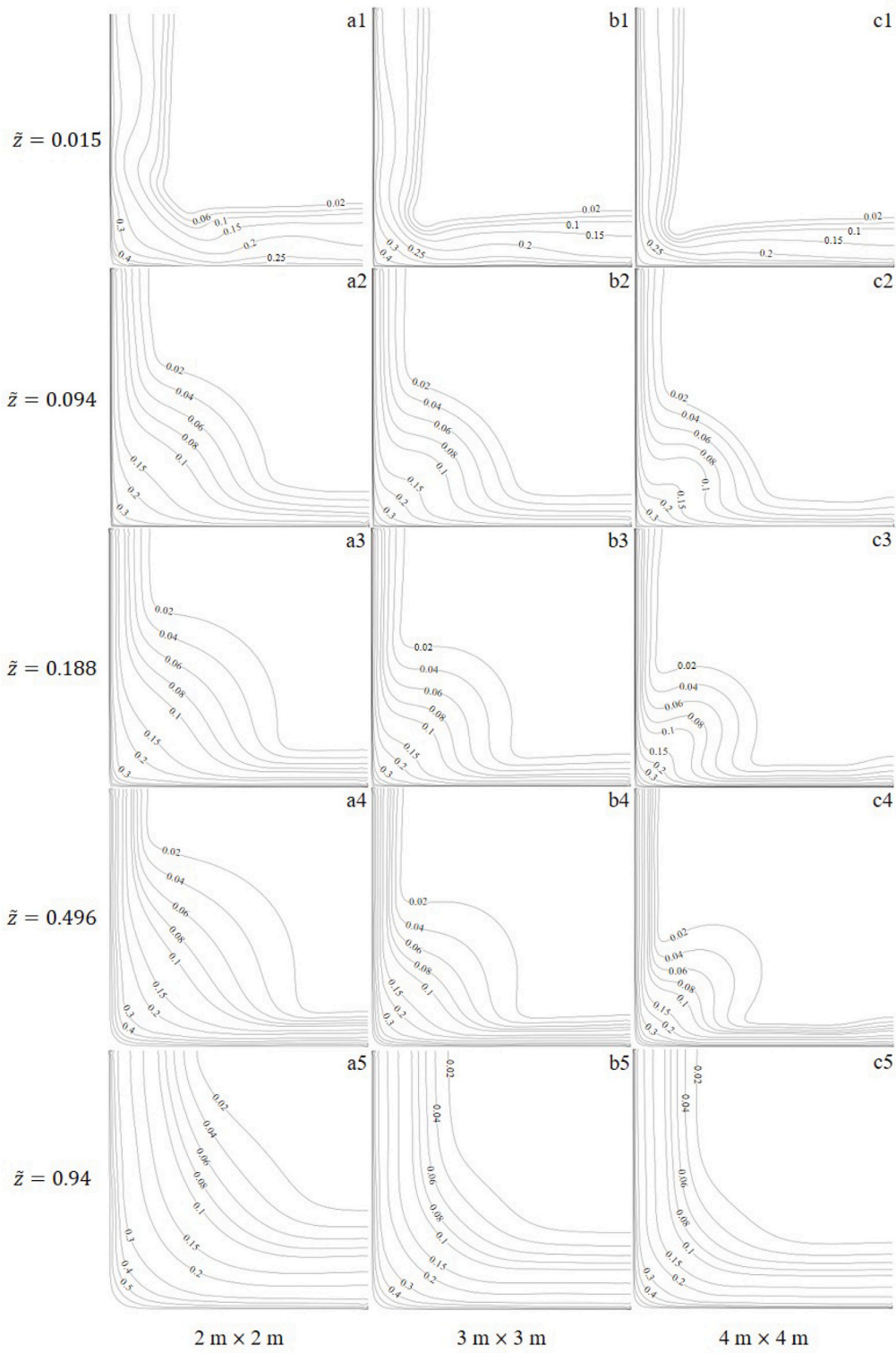
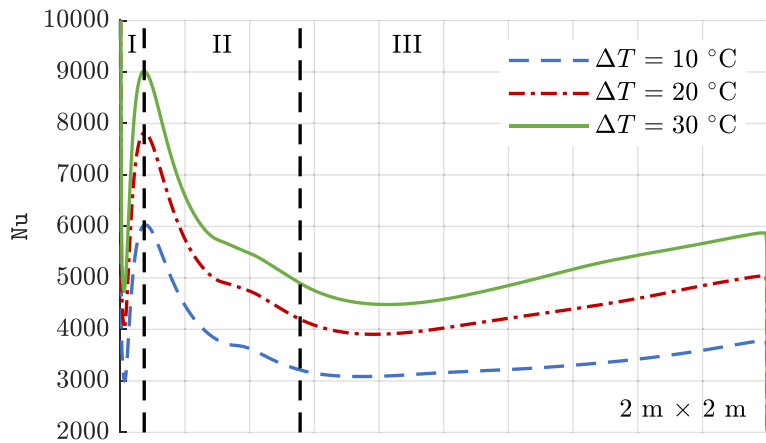
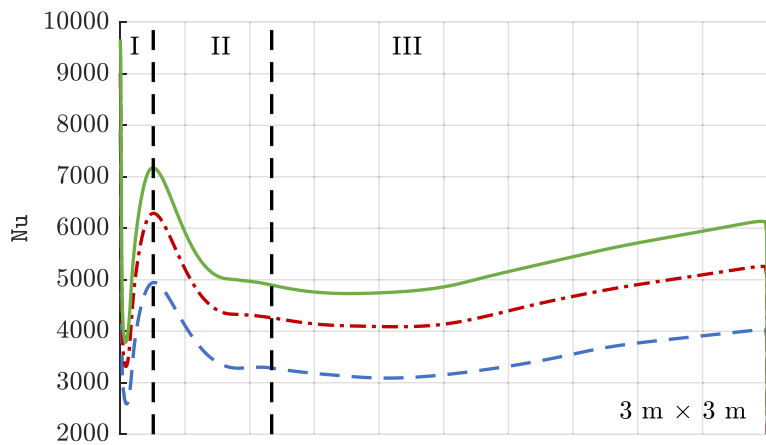


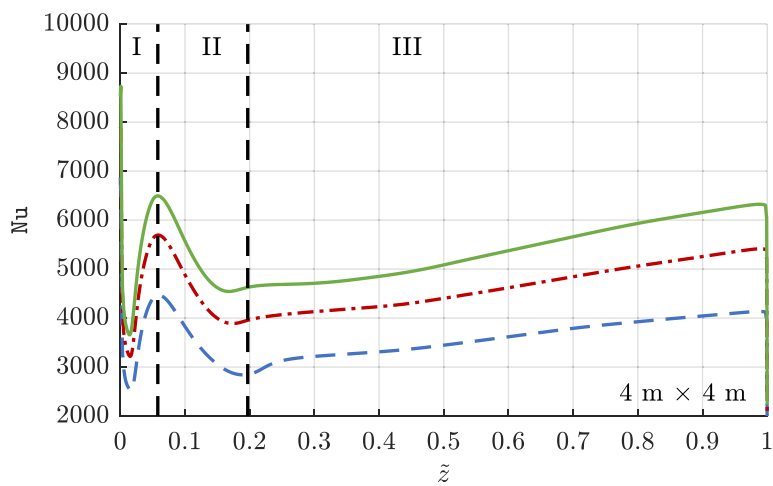
Fig. 13. Impact of the shaft width on the isotherm plots at different cross-sectional planes in the shaft for a temperature difference of 10°C. Contour values are non-dimensionalized according to T^* in Table 2. Shaft width (a1–a5) $b = 2$ m, (b1–b5) $b = 3$ m, and (c1–c5) $b = 4$ m.



(a)



(b)



(c)

Fig. 14. Impact of temperature difference and shaft width on the Nusselt number profile along $\tilde{z}$ for (a) $b = 2$ m, (b) $b = 3$ m and (c) $b = 4$ m.

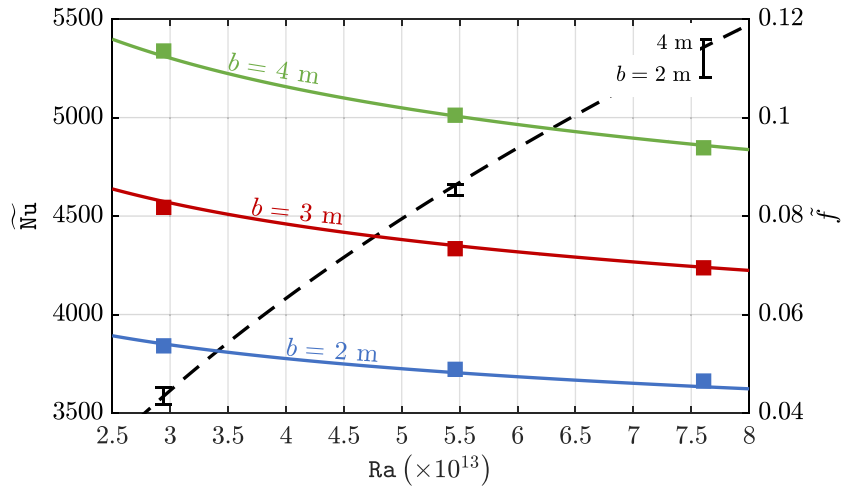


Fig. 15. Variation of the average Nusselt Number and the friction factor with Rayleigh's number.

$$\widetilde{Nu} = 0.0072 Ra^{0.4231} \quad (24)$$

$$\tilde{f} = 314.72 \left(\frac{b}{H} \right)^{1.0576} Ra^{-0.185} \quad (25)$$

where $\ell_1^* = \ell_1/H$ and $\ell_2^* = \ell_2/H$. We propose these building-scale correlations as practical tools for roughly predicting flow behavior during the preliminary design of buoyancy-driven ventilation shafts. The conditions under which Eqs. (24)–(25) could be applied are as follows:

$$\left[\begin{array}{c} Pr = 0.71 \\ 2.94 \times 10^{13} \leq Ra \leq 7.61 \times 10^{13} \end{array} \right]$$

4. Conclusion

This study explores employing buoyancy-driven flows, established by isothermally heating a hollow shaft that runs through the core of a building, to ventilate urban street canyons at the pedestrian level. Three-dimensional numerical simulations are conducted in ANSYS-Fluent using the SST $k-\omega$ model to account for turbulence effects. A parametric study is conducted to assess the impact of the geometry and thermal conditions on the ventilation performance. The parameters varied are the shaft width {2 m, 3 m, 4 m} and the temperature difference between the walls and the ambient {10 °C, 20 °C, 30 °C}. The impact of far-field boundary conditions was also assessed by comparing cases in which the domain boundaries were modeled either as stationary walls or as zero gauge pressure outlets. The relative differences between these two setups do not exceed 10% for the ventilation mass flow rate and 1.5% for the wall heat flux. The resulting mass flow rate, for the far-field wall configuration, ranges between 3.76 and 16.63 kg/s, whereas the ventilation effectiveness ranges between 0.179 and 0.645 kg/kJ. In all cases, the flow is found to be in the developing regime, exhibiting a thermal behavior analogous to that of four independent isothermally heated vertical plates. In the absence of experimental data, partial validation of the results is achieved by modeling the buoyancy-driven flow over a flat plate using the same numerical setup as that employed in the three-dimensional simulations; the numerical predictions agree with empirical results of an isothermally-heated flat plate within 15%. Moreover, the numerical simulations showed a qualitative agreement with the predictions of simplified models established for buoyancy-driven flows. A notable improvement in the predictions of these models was achieved when the friction factor predicted by the CFD simulations, rather than empirical estimates for pressure driven flows, is used. The three-dimensional nature of the flow is observed by plotting the streamlines, which show flow separation in all cases and delayed reattachment in wider shafts. Inspection of the Nu profile, the axial velocity profiles, and the isotherm plots shows that the flow in the shaft is divided into three zones: recirculation zone, core flow zone, and boundary-dominated zone. Based on the numerical results, we establish building-scale correlations for the average Nusselt number $\widetilde{Nu}$, average friction factor $\tilde{f}$, the entry length ℓ_1 and the core flow length ℓ_2 as functions of Rayleigh number Ra and the shaft aspect ratio (b/H). We expect this study to serve as the starting point for future studies that explore the impact on the ventilation capacity and thermal comfort indices of the shaft height and shape, in addition to feasible and cost-effective means of establishing the desired temperature difference.

Nomenclature

Nu	Nusselt number	K	Coefficient of minor losses
Gr	Grashof number	D_h	Hydraulic diameter
Pr	Prandtl number	q''	Local heat Flux
Ra	Rayleigh number	A	Shaft horizontal cross-sectional area
Re	Reynolds number	Greek Symbols	
f	Friction factor	ρ	Density
b	Shaft width	λ	Thermal conductivity
H	Shaft height	μ	Dynamic viscosity
L	Characteristic length	β	Thermal expansion coefficient
$\vec{u}$	Velocity vector	Φ	Viscous dissipation
g	Gravitational acceleration	ε	Dissipation rate of k
T	Temperature	ω	Specific turbulent dissipation rate
p	Pressure	Γ_k	Effective diffusivity of k
R	Ideal gas constant	Γ_ω	Effective diffusivity of ω
c_p	Constant pressure specific heat	$\zeta_{\max}$	Maximum aspect ratio in the grid
h	Convection heat transfer coefficient	δ_{th}	Thermal boundary layer thickness
(x, y, z)	Cartesian coordinates	Π	Perimeter
$\hat{z}$	Unit vector in the z direction	ν	Kinematic viscosity
k	Turbulent kinetic energy	α	Thermal diffusivity
ω	Specific turbulence dissipation	τ_w	Wall shear stress
G_k	Production of k	Subscripts	
Y_k	Dissipation of k	w	Wall
S_k	Source term of k	∞	Ambient
G_b	Turbulence generation of k due to buoyancy	t	Turbulent
		i	Inlet
G_ω	Production of ω	o	Outlet
Y_ω	Dissipation of ω	f	Film
S_ω	Source term of ω	num	Numerical
$G_{\omega b}$	Turbulence generation of ω due to buoyancy	emp	Empirical
		z	z -component of a vector
D_ω	Cross-diffusion term	tot	Total
t_1	First inflation layer thickness	Superscripts	
Δx	Grid size in the shaft	$*$	Non-dimensional form of a variable
		Diacritic	
N_E	Number of grid elements	—	Mean value based on empirical data
		$\sim$	Mean value based on numerical data
y^+	Normalized distance to a wall	Abbreviations	
$\dot{m}$	Mass flow rate	CFD	Computational Fluid Dynamics
$\dot{Q}$	Wall heat transfer rate	RANS	Reynolds Average Navier Stokes
$\bar{z}$	Elevation normalized by shaft height		

CRedit authorship contribution statement

Maram Ammar: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis. **Charbel Ziade:** Software, Validation, Writing – review & editing. **Aram Yeretizian:** Writing – original draft, Methodology, Investigation. **Fadi Moukalled:** Writing – original draft, Software, Methodology, Investigation, Formal analysis. **Issam Lakkis:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Methodology, Investigation, Formal analysis, Conceptualization.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used ChatGPT and Deepseek in order to improve language and readability. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

Funding sources

This research was supported by the University Research Board of the American University of Beirut: Project number 28042, Award number: 104632

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The authors would like to acknowledge Mher Kazandjian for his support in running the simulations using HPC resources at AUB. The authors would also like to acknowledge Shareef Tarhini and the Beirut Urban Lab for providing us with the data on building statistics in Beirut.

Appendix A. Coefficients of the SST $k - \omega$ model

Transport Equation of the Turbulence Kinetic Energy:

$$\nabla \cdot (\rho \vec{u} k) = \nabla \cdot (\Gamma_k \nabla k) + G_k - Y_k + S_k + G_b \quad (\text{A.1})$$

where G_k is the production of turbulence kinetic energy, Y_k is the dissipation of k , S_k is a user-defined source term and G_b accounts for buoyancy.

Transport Equation of the Specific Dissipation Rate:

$$\nabla \cdot (\rho \vec{u} \omega) = \nabla \cdot (\Gamma_\omega \nabla \omega) + G_\omega - Y_\omega + D_\omega + S_\omega + G_{b\omega} \quad (\text{A.2})$$

where G_ω is the generation of turbulence kinetic energy, Y_ω is the dissipation of ω , S_ω is a user-defined source term and $G_{b\omega}$ accounts for buoyancy.

where

$$\rightarrow \Gamma_k = \frac{1}{\frac{F_1}{\sigma_{k,1}} + \frac{1-F_1}{\sigma_{k,2}}} \quad (\text{A.3})$$

$$\rightarrow \Gamma_\omega = \frac{1}{\frac{F_1}{\sigma_{\omega,1}} + \frac{1-F_1}{\sigma_{\omega,2}}} \quad (\text{A.4})$$

$$F_1 = \tanh \Phi_1^4 \quad (\text{A.5})$$

$$\Phi_1 = \min \left[\max \left(\frac{\sqrt{k}}{0.09\omega y}, \frac{500\mu}{\rho y^2 \omega} \right), \frac{4\rho k}{\sigma_{\omega,2} D_\omega^+ y^2} \right] \quad (\text{A.6})$$

where y is the distance to the next surface

$$D_\omega^+ = \max \left[2\rho \frac{1}{\sigma_{\omega,2}} \frac{1}{\omega} \nabla k \cdot \nabla \omega, 10^{-10} \right] \quad (\text{A.7})$$

$$\sigma_{k,1} = 1.176 \quad \sigma_{k,2} = 1$$

$$\sigma_{\omega,1} = 2.0 \quad \sigma_{\omega,2} = 1.168$$

$$\rightarrow G_k = \mu_t S^2 \quad \text{where} \quad S = \sqrt{2(\mathbf{S} : \mathbf{S})} \quad (\text{A.8})$$

$$\mathbf{S} = \frac{1}{2} \left(\nabla \vec{u} + (\nabla \vec{u})^T \right) \quad (\text{A.9})$$

$$\rightarrow G_\omega = \frac{\alpha \alpha^*}{v_t} G_k \quad (\text{A.10})$$

$$\alpha = \frac{\alpha_\infty}{\alpha^*} \left(\frac{\alpha_0 + \frac{Re_t}{R_\omega}}{1 + \frac{Re_t}{R_\omega}} \right) \quad (\text{A.11})$$

$$\alpha^* = \alpha_\infty^* \left(\frac{\alpha_0^* + \frac{Re_t}{R_k}}{1 + \frac{Re_t}{R_k}} \right) \quad (\text{A.12})$$

$$Re_t = \frac{\rho k}{\mu \omega} \quad (\text{A.13})$$

$$R_k = 6 \quad R_\omega = 2.95 \quad \alpha_0 = \frac{1}{9} \quad \alpha_\infty^* = 1 \quad \alpha_0^* = 0.024$$

$$\alpha_\infty = F_1 \alpha_{\infty,1} + (1 - F_1) \alpha_{\infty,2} \quad (\text{A.14})$$

$$\alpha_{\infty,1} = \frac{\beta_{i,1}}{\beta_\infty^*} - \frac{\kappa^2}{\sigma_{\omega,1} \sqrt{\beta_\infty^*}} \quad (\text{A.15})$$

$$\alpha_{\infty,2} = \frac{\beta_{i,2}}{\beta_\infty^*} - \frac{\kappa^2}{\sigma_{\omega,2} \sqrt{\beta_\infty^*}} \quad (\text{A.16})$$

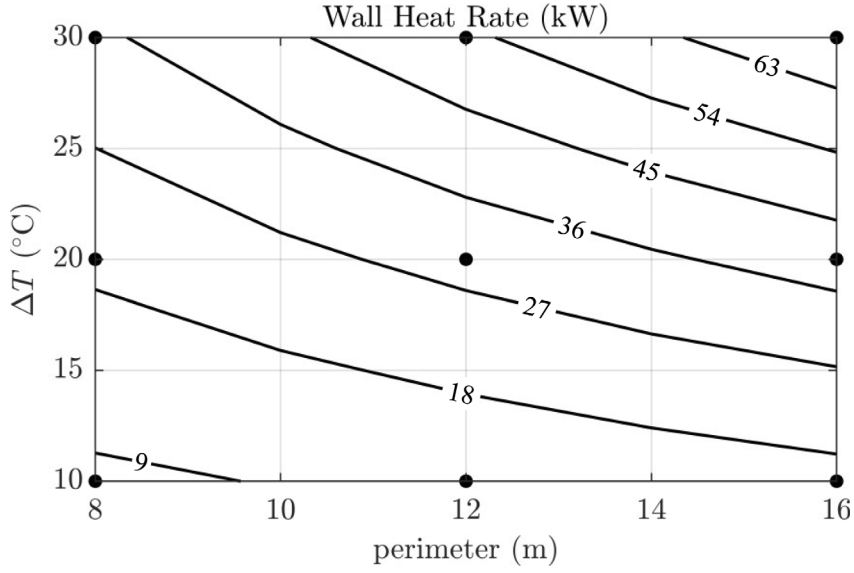


Fig. B.1. Wall heat flux as a function of shaft width and temperature difference. Perimeter = 4b.

$$\kappa = 0.41 \quad \beta_{i,1} = 0.075 \quad \beta_{i,2} = 0.0828 \quad \beta_{\infty}^* = 0.09$$

$$\rightarrow Y_k = \rho \beta^* k \omega \quad (\text{A.17})$$

$$\text{For incompressible flows: } \beta^* = \beta_i^* = \beta_{\infty}^* \left(\frac{\frac{4}{15} + \left(\frac{Re_t}{R_{\beta}} \right)^4}{1 + \left(\frac{Re_t}{R_{\beta}} \right)^4} \right) \quad (\text{A.18})$$

$$\beta_{\infty}^* = 0.09 \quad R_{\beta} = 8 \quad (\text{A.19})$$

$$\rightarrow Y_{\omega} = \rho \beta_c \omega^2 \quad (\text{A.20})$$

β_c is a constant

For incompressible flows: $\beta_c = \beta_i = F_1 \beta_{i,1} + (1 - F_1) \beta_{i,2}$

$$\rightarrow D_{\omega} = 2(1 - F_1) \rho \frac{1}{\omega \sigma_{\omega,2}} \nabla k \cdot \nabla \omega \quad (\text{A.21})$$

$$\rightarrow G_b = \frac{\mu_t}{\rho Pr_t} \vec{g} \cdot \nabla \rho \quad \text{for an ideal gas} \quad (\text{A.22})$$

$$Pr_t = 0.85$$

$$G_{\omega b} = \frac{\omega}{k} [(1 + \alpha) C_{3\epsilon} G_b - G_b]$$

$$C_{3\epsilon} = \tanh \left(\left| \frac{v}{u} \right| \right) \quad (\text{A.23})$$

where $\vec{g}$ is the gravitational vector field, v is the component of the velocity parallel to $\vec{g}$ and u is the component of the velocity perpendicular to $\vec{g}$

Appendix B. Additional figures

See Fig. B.1.

Appendix C. Derivation of the simplified models

C.1. Stack model

The net buoyancy pressure difference (Δp_b) between the shaft inlet and exit equal to the hydrostatic imbalance, as expressed in Eq. (C.24).

$$\Delta p_b = \int_0^H (\rho_\infty(z) - \rho(z))g dz \quad (C.24)$$

where $\rho(z)$ is the air density at elevation z inside the shaft and $\rho_\infty(z)$ is the ambient density taken to be constant. We assume that the inlet density (ρ_i) is equal to the ambient density and that the temperature difference between the ambient air and the stack air permits invoking the Boussinesq approximation for an ideal gas. This approximation linearly relates the temperature to the density, rendering Eq. (C.24) as follows:

$$\Delta p_b = \int_0^H \rho_i \beta (T(z) - T_i) g dz \quad (C.25)$$

where β is the thermal expansion coefficient. To simplify this model, we further assume that the air temperature in the stack increases linearly from T_i at the entrance ($z = 0$) to T_o at the exit ($z = H$), which yields

$$\Delta p_b = \rho_i g H \beta \frac{\Delta T}{2} \quad (C.26)$$

where $\Delta T = T_o - T_i$.

The rate of heat transfer from the shaft walls to the stack air $\dot{Q}^+$ is estimated by applying the energy equation under steady state conditions as follows:

$$\dot{Q}^+ = \dot{m} c_p \Delta T \quad (C.27)$$

We now consider two scenarios: (i) Ideal model where the buoyancy pressure difference is converted to dynamic pressure ($0.5 \rho_i |\vec{u}|^2$) without any losses. (ii) Non-ideal model, which is more realistic since it accounts for losses at the entrance, exit, and walls during the conversion of Δp_b into dynamic pressure.

(i) Ideal Model

$$\begin{aligned} \Delta p_b &= \frac{1}{2} \rho_i u^2 \\ u_{\text{stack, ideal}} &= \sqrt{g H \beta \Delta T} \end{aligned} \quad (C.28)$$

$$\dot{m}_{\text{stack, ideal}} = \rho_i u_{\text{stack, ideal}} A = \sqrt{\rho_i^2 A^2 g H \beta \Delta T} = \left(\frac{\rho_i^2 A^2 g H \beta \dot{Q}^+}{c_p} \right)^{\frac{1}{3}} \quad (C.29)$$

(ii) Non-ideal Model

$$\begin{aligned} \Delta p_b &= \frac{(1 + K_{\text{tot}})}{2} \rho_i u^2 \\ u_{\text{stack, loss}} &= \sqrt{\frac{g H \beta \Delta T}{(1 + K_{\text{tot}})}} \end{aligned} \quad (C.30)$$

$$\dot{m}_{\text{stack, loss}} = \rho_i u_{\text{stack, loss}} A = \sqrt{\frac{\rho_i^2 A^2 g H \beta \Delta T}{1 + K_{\text{tot}}}} = \left(\frac{\rho_i^2 A^2 g H \beta \dot{Q}^+}{c_p (1 + K_{\text{tot}})} \right)^{\frac{1}{3}} \quad (C.31)$$

$$K_{\text{tot}} = K_{\text{in}} + K_{\text{out}} + \frac{\bar{f} H}{D_h} \quad (C.32)$$

where u is the mean speed (m/s), $\dot{m}$ is the mass flow rate (kg/s), A is the cross-sectional area of the shaft (m²), K_{tot} is the total coefficient of losses due to the inlet, outlet and friction at walls, D_h is the hydraulic diameter, and $\bar{f}$ is the Darcy friction factor computed using Swamee-Jain's formula expressed in Eq. (C.33).

$$\bar{f} = \frac{0.25}{\left[\log_{10} \left(\frac{\epsilon}{3.7 D_h} + \frac{5.74}{\text{Re}_{D_h}^{0.9}} \right) \right]^2}, \quad (C.33)$$

Note that the expression of the mass flow rate in the ideal case can be recovered by setting and $K_{\text{tot}} = 0$.

For this scenario, we can establish upper and lower bounds for the mass flow rate induced in the shaft by considering two limiting conditions. The first limiting condition gives the upper bound ($\dot{m}_{\text{stack, losses}}^{\dagger}$) and assumes that the stack air

temperature reaches the wall temperature rapidly. In other words, the convection heat transfer coefficient at the walls is assumed to be infinite in this case, and the wall heat rate is given by Eq. (C.27). Thus,

$$\dot{m}_{\text{stack, losses}}^{\uparrow} = \sqrt{\frac{\rho_i^2 A^2 g H \beta \Delta T}{(1 + K_{\text{tot}})}} \quad (\text{C.34})$$

The second limiting condition gives the lower bound and assumes weak heating, where the stack air temperature is taken to be equal to the inlet temperature over the shaft height. In this limiting case, the convection heat transfer is finite, and the heat rate of the walls can be computed as follows:

$$\dot{Q} = \bar{h} \Pi H (T_w - T_i) \quad (\text{C.35})$$

$$\text{Nu} = \frac{\bar{h} H}{\lambda} \quad (\text{C.36})$$

where Π is the shaft perimeter (m), $\bar{h}$ is the average convection heat transfer coefficient ($\text{W/m}^2 \text{K}$), λ is the thermal conductivity of air (W/mK), and Nu is the average Nusselt number computed for a vertical flat plate according to Eq. (19). As such, the lower bound on the mass flow rate in the shaft is

$$\dot{m}_{\text{stack, losses}}^{\downarrow} = \left(\frac{\rho_i^2 A^2 g H \beta}{c_p (1 + K_{\text{tot}})} \bar{h} \Pi H (T_w - T_i) \right)^{\frac{1}{3}} \quad (\text{C.37})$$

In the three-dimensional simulations, the outlet temperature remains close to the ambient value, indicating that the second limiting condition (weak heating) provides a more accurate representation of the flow in the shaft. Accordingly, in Section 2.5.1, we report $\dot{m}_{\text{stack, losses}}^{\downarrow}$ as the mass flow rate predicted by the stack model. We note here that the coefficient of total losses K_{tot} depends on the friction factor, which itself is a function of Reynolds number. However, in buoyancy-driven flows, Re is not known beforehand, which requires an iterative procedure to estimate the mass flow rate. Moreover, $\bar{h}$ is computed from Nusselt number correlations for isothermally heated vertical plates. As such, it is independent of the mass flow rate in this formulation.

C.2. One-dimensional model

In this section, we present a simplified one-dimensional model to roughly estimate the mass flow rate, wall heat rate, and other characteristics of the buoyancy-driven flow established in the shaft. Even though this model embeds many assumptions, it gives a more detailed description of the flow than the stack model. We consider an isothermally-heated vertical shaft, with a height H , cross-sectional area A , and a perimeter Π . Air is modeled as an ideal gas with constant values for the specific heat c_p (J/kg K), thermal conductivity λ (W/m K), and ideal gas constant R (J/kg K). The outlet ($z = H$) is taken to be open to the ambient, where the inlet and outlet static pressures are related to the ambient conditions at the corresponding elevations by taking into account the dynamic head and inlet/outlet losses.

$$p(H) = p_o = p_i - \rho_i g H \quad (\text{C.38})$$

where subscripts i and o denote inlet and outlet, respectively. By the ideal gas law,

$$p(z) = \rho(z) R T(z) \quad (\text{C.39})$$

We apply the integral form of the conservation laws of mass, momentum and energy, under steady state conditions, to a control volume having a cross-sectional area A and extending from z to $z + dz$. We, then, check the validity of the model using the entropy balance equation.

C.2.1. Conservation of mass

$$\dot{m} = \rho(z) u(z) A = \text{cst} \quad \Rightarrow \quad u(z) = \frac{\dot{m}}{\rho(z) A} \quad (\text{C.40})$$

where $u = u(z)$ is the axial velocity

C.2.2. Conservation of momentum

$$\rho u \frac{du}{dz} = -\frac{dp}{dz} - \rho g - f_{\tau} \quad (\text{C.41})$$

where f_{τ} is the shear force per unit volume expressed as follows:

$$f_{\tau} = \frac{\tau_w \Pi dz}{A dz} = \frac{\tau_w \Pi}{A} = \frac{4\tau_w}{D_h} \quad (\text{C.42})$$

where, τ_w is the wall shear stress (N/m²), Π is the shaft perimeter (m), and $D_h = (4A)/\Pi$ is the hydraulic diameter (m). The shear stress at the wall can be modeled as follows:

$$\tau_w = \frac{f \rho u^2}{8} \quad (\text{C.43})$$

where f is the friction factor computed according to Eq. (C.33). The momentum equation then becomes,

$$\rho u \frac{du}{dz} = -\frac{dp}{dz} - \rho g - \left(\frac{f \rho u^2}{2D_h} \right) \quad (\text{C.44})$$

To simplify the analysis, we will assume that the momentum flux ($\rho u(du/dz)$) is negligible and subtract the hydrostatic conditions, expressed in Eq. (C.45), from the momentum equation, resulting in Eq. (C.46).

$$\frac{dp_\infty}{dz} = -\rho_\infty g \quad (\text{C.45})$$

$$\frac{d(p - p_\infty)}{dz} = -(\rho - \rho_\infty)g - \frac{f \rho u^2}{2D_h} \quad (\text{C.46})$$

We observe in Eq. (C.46) that the buoyancy term $((\rho_\infty - \rho)g)$, balances the friction losses $(f \rho u^2)/2D_h$ and the minor losses $((d(p - p_\infty))/dz)$. Integrating Eq. (C.46) over the shaft height yields

$$\int_0^H (\rho_\infty - \rho(z))g A dz = \int_0^H \frac{f \rho u^2}{8} \Pi dz + \int_0^H d(p - p_\infty) A \quad (\text{C.47})$$

where the friction losses in addition to the inlet and outlet losses are denoted by Δp_{loss} , presented in the following equation:

$$\begin{aligned} \Delta p_{\text{loss}}(\dot{m}) &= \frac{1}{A} \int_0^H \frac{f \rho u^2}{8} \Pi dz + \frac{1}{A} \int_0^H d(p - p_\infty) \\ &= \frac{1}{2} \rho_* u^2 \left(K_i + K_o + \bar{f} \frac{H}{D_h} \right) \\ &= \frac{1}{2} \frac{\dot{m}^2}{\rho_* A^2} \left(K_i + K_o + \bar{f} \frac{H}{D_h} \right) \end{aligned} \quad (\text{C.48})$$

where ρ_* is evaluated at the bulk mean temperature, $u = \dot{m}/(\rho_* A)$, $K_i = 0.5$, and $K_o = 1.0$ are the coefficients of minor losses of a sharp-edged entrance and exit respectively. Note that Eq. (C.47) is obtained by integrating Eq. (C.46) over the volume, so the friction term becomes $\int_0^H f_\tau A dz$ which is the same as $\int_0^H \tau_w \Pi dz$.

The buoyancy stack is denoted by Δp_{stack} , and it is presented in Eq. (C.49)

$$\Delta p_{\text{stack}}(\dot{m}) = \int_0^H g(\rho_\infty - \rho) dz = g \rho_\infty \int_0^H \left(1 - \frac{T_\infty}{T(z)} \right) dz \quad (\text{C.49})$$

where $\rho_\infty = p_\infty/(RT_\infty)$ and $T(z)$ is determined from the energy equation, presented in the following section. We note here that in Eq. (C.49), we assumed that the pressure in the shaft is close to the ambient pressure such that $\rho(z) = p_\infty/(RT(z))$, where $T(z)$ is a function of $\dot{m}$. Finding the mass flow rate in the shaft, then, consists of balancing the friction and minor losses with the buoyancy pressure stack $\Rightarrow \Delta p_{\text{stack}} = \Delta p_{\text{loss}}$

C.2.3. Conservation of energy

$$\dot{m} \left(\frac{d\tilde{h}}{dz} + u \frac{du}{dz} + g \right) dz = q^- \Pi dz \quad (\text{C.50})$$

where $\tilde{h}$ is the specific enthalpy (kJ/kg), and q^- is the wall heat flux (W/m²). In Eq. (C.50), we neglected heat transfer by conduction in the axial direction, which could be justified by the low conductivity of air. Combining Eq. (C.44) with Eq. (C.50) yields

$$\frac{d\tilde{h}}{dz} = \frac{q^- \Pi}{\dot{m}} + \frac{1}{\rho} \frac{dp}{dz} + \frac{f u^2}{2D_h} \quad (\text{C.51})$$

Since air is modeled as an ideal gas, it follows that $d\tilde{h}/dz = c_p(dT/dz)$, which yields the following form of the energy equation,

$$\frac{dT}{dz} = \frac{q^- \Pi}{\dot{m} c_p} + \frac{1}{\rho c_p} \frac{dp}{dz} + \Phi \quad (\text{C.52})$$

where $\Phi = (f u^2)/(2c_p D_h)$ is the viscous dissipation rate per unit length. We further simplify our model by assuming that Φ and $(1/\rho c_p)(dp/dz)$ are negligible, which gives

$$\frac{dT}{dz} = \frac{q^- \Pi}{\dot{m} c_p} \quad (\text{C.53})$$

The wall heat flux q^- is estimated according to the following equation:

$$q^- = \bar{h} (T_w - T(z)) \quad (\text{C.54})$$

where the average heat transfer coefficient, $\bar{h}$, is computed according to Eq. (C.36) and Eq. (19). Since $\dot{m}$ is constant and $\bar{h}$ is assumed to be constant, integrating Eq. (C.53) yields

$$T(z) = T_w - (T_w - T_i) \exp\left(-\frac{\bar{h}\Pi z}{\dot{m}c_p}\right) \quad (\text{C.55})$$

The average bulk temperature used to compute ρ_* in Eq. (C.48) is given in the following equation:

$$\bar{T} = \frac{1}{H} \int_0^H T(z) dz = T_w - (T_w - T_i) \frac{1 - \exp\left(-\frac{\bar{h}\Pi z}{\dot{m}c_p}\right)}{(H\bar{h}\Pi)/(\dot{m}c_p)} \quad (\text{C.56})$$

C.2.4. Entropy balance

In this section, we prove that the simplified one-dimensional model does not violate the second law of thermodynamics, which constrains the generated entropy to be positive. Applying the entropy balance to the selected control volume yields the following equation.

$$\dot{m} \frac{ds}{dz} = \frac{q^* \Pi}{T_w} + \sigma_{\text{gen}} \quad (\text{C.57})$$

where σ_{gen} is the rate of entropy generated in the control volume per unit length. From Gibbs' relation, we know that

$$T ds = d\tilde{h} - \frac{dp}{\rho} \quad (\text{C.58})$$

For an ideal gas, $d\tilde{h} = c_p dT$. Hence,

$$\dot{m} T \frac{ds}{dz} = \dot{m} c_p \frac{dT}{dz} - \frac{\dot{m}}{\rho} \frac{dp}{dz} \quad (\text{C.59})$$

Substituting $\dot{m} c_p \frac{dT}{dz}$ with its expression from Eq. (C.52) in Eq. (C.59) gives

$$\dot{m} \frac{ds}{dz} = \frac{q^* \Pi}{T} + \frac{\dot{m} c_p \Phi}{T} \quad (\text{C.60})$$

By taking the difference between Eqs. (C.57) and (C.60), σ_{gen} becomes

$$\sigma_{\text{gen}} = q^* \Pi \left(\frac{1}{T} - \frac{1}{T_w} \right) + \frac{\dot{m} f u^2}{2 D_h T} \geq 0 \quad (\text{C.61})$$

Similarly to the stack model, the mass flow rate is determined in an iterative procedure, since the friction factor, f , depends on Reynolds number, which is not known a priori. The solution procedure followed to compute $\dot{m}$ is outlined in Algorithm 1

Algorithm 1 Iterative method for computing $\dot{m}$ from the one-dimensional model

- 1: Initialize $\dot{m}$ and define a tolerance ϵ_{tol}
 - 2: **repeat**
 - 3: Find f from forced flow correlations (Equation (C.33))
 - 4: Calculate $\bar{h}$ from Equation (18) and Equation (19)
 - 5: Compute $T(z)$ from Equation (C.56)
 - 6: Compute $\rho(z)$ from the ideal gas law and $u(z)$ from Equation (C.40)
 - 7: Compute the buoyancy stack by integrating Equation (C.49) over H
 - 8: Compute minor losses and friction losses from Equation (C.48)
 - 9: **if** $|\Delta p_{\text{stack}} - \Delta p_{\text{loss}}| \geq \epsilon_{\text{tol}}$ **then**
 - 10: Update $\dot{m}$
 - 11: **end if**
 - 12: **until** Convergence
 - 13: Calculate the total rate of heat transfer at the walls $\dot{Q} = \bar{h}\Pi H(T_w - T_\infty)$
-

Data availability

Data will be made available on request.

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